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(54) **CAMERA OPTICAL LENS**

(52) **U.S. Cl.**

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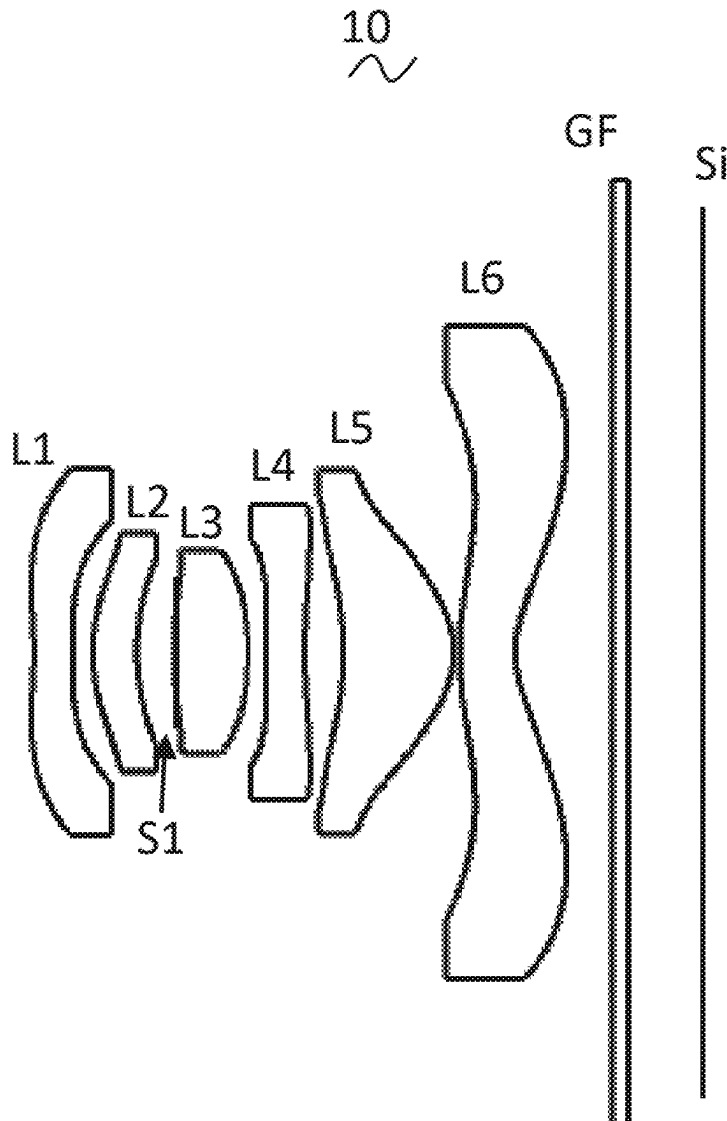
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G02B 13/18 (2006.01)

(57) **ABSTRACT**

The present disclosure relates to the field of optical lens, and provides a camera optical lens, including, from an object side to an image side, a first lens, a second lens, a third lens, a fourth lens, a fifth lens and a sixth lens. The camera optical lens satisfies: $6.00 \leq R2/R1 \leq 25.00$, $2.00 \leq (R7+R8)/(R7-R8) \leq 9.00$, and $3.00 \leq (d5+d9)/d7 \leq 9.00$. R1 represents a curvature radius of an object side surface of the first lens, R2 represents a curvature radius of an image side surface of the first lens, R7 represents a curvature radius of an object side surface of the fourth lens, R8 represents a curvature radius of an image side surface of the fourth lens, d5 represents an on-axis thickness of the third lens, d7 represents an on-axis thickness of the fourth lens, and d9 represents an on-axis thickness of the fifth lens.



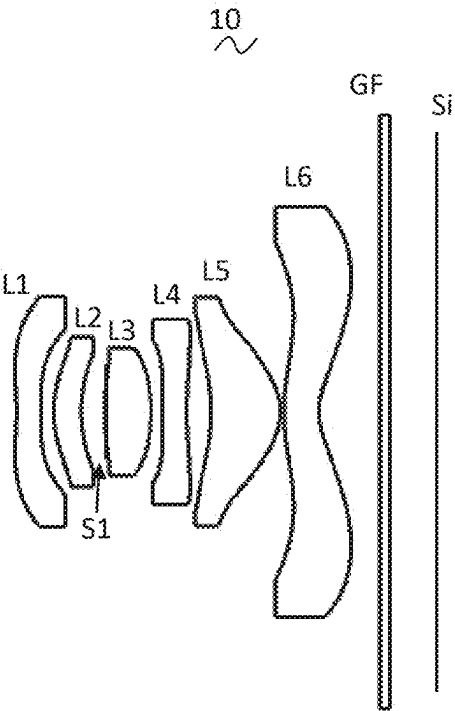


FIG. 1

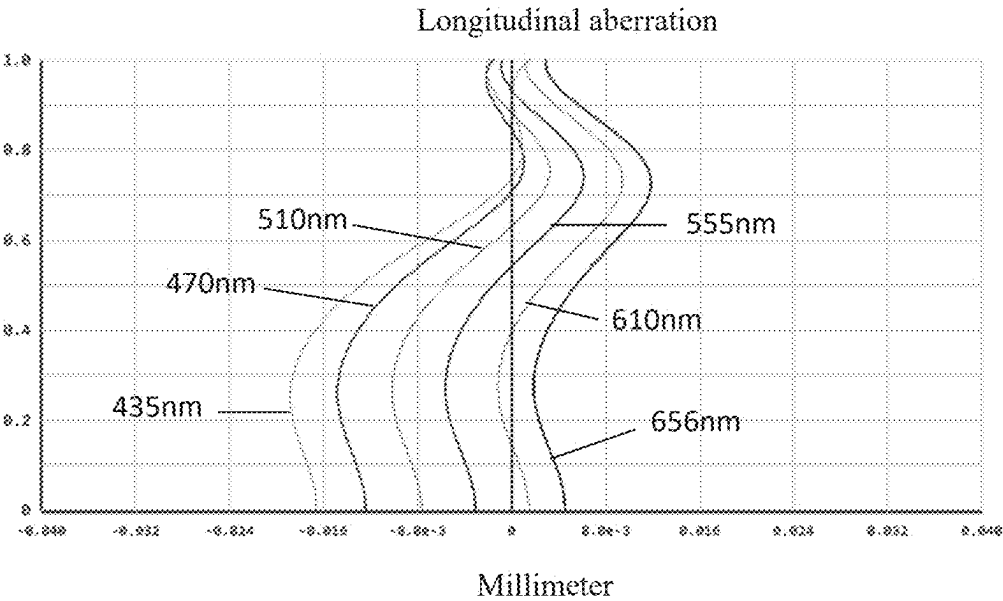


FIG. 2

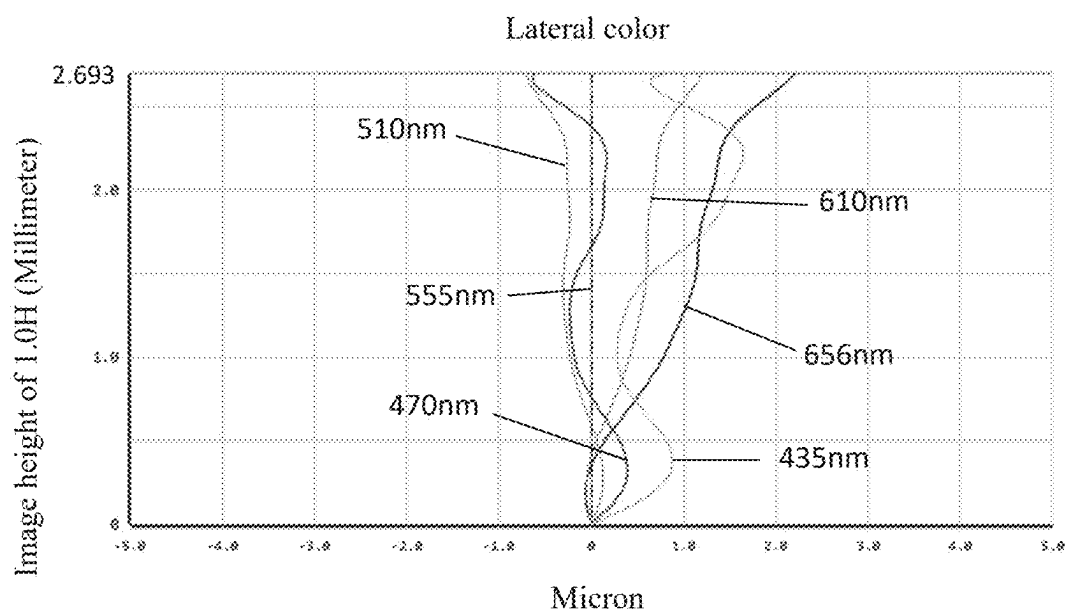


FIG. 3

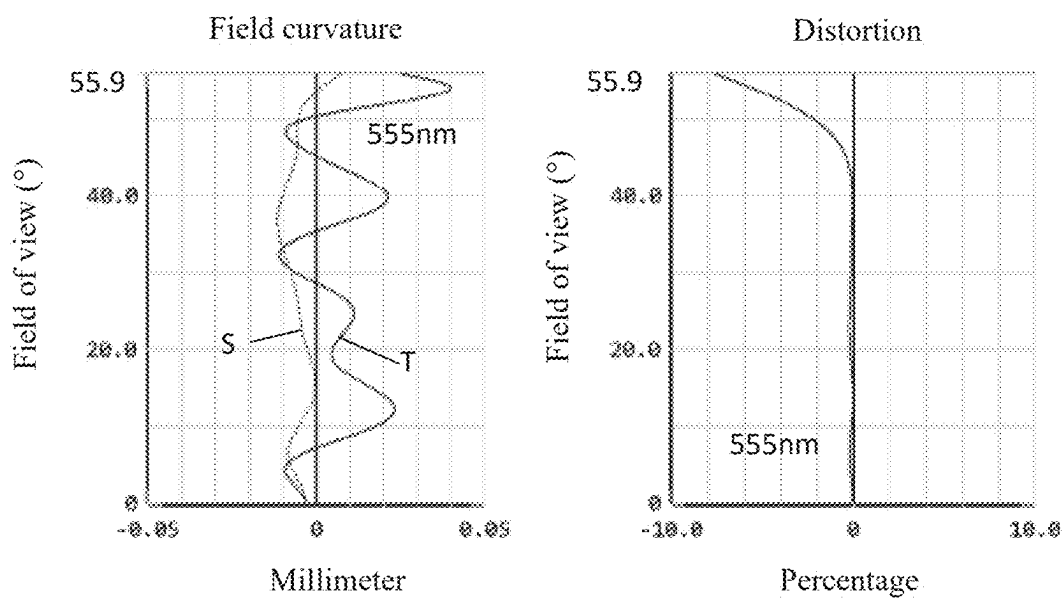


FIG. 4

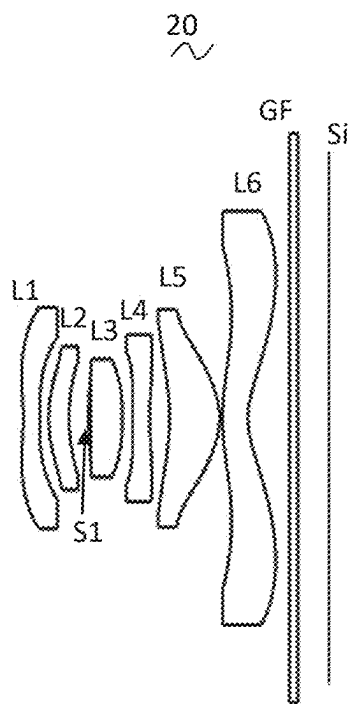


FIG. 5

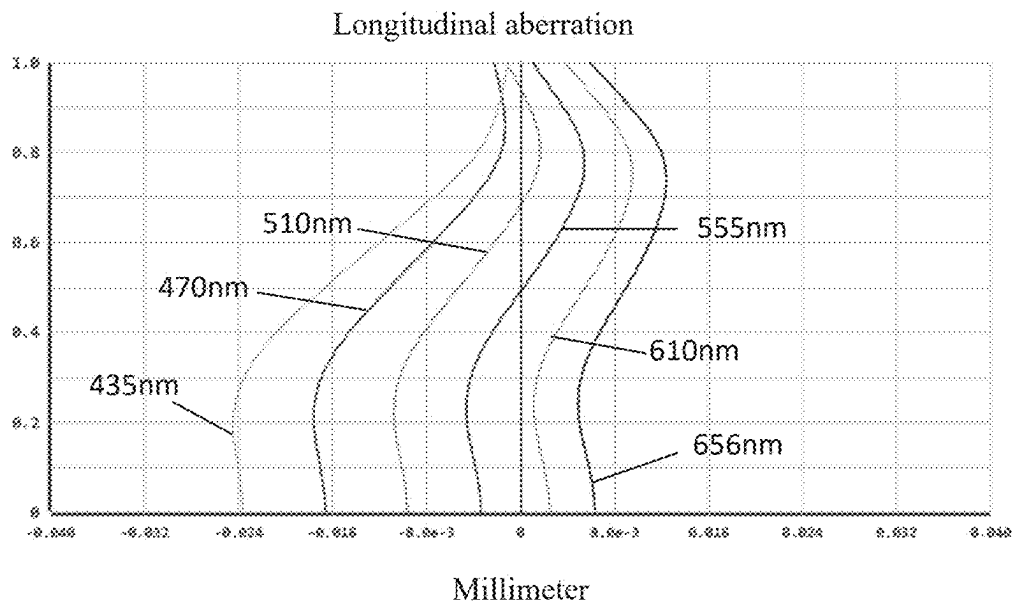


FIG. 6

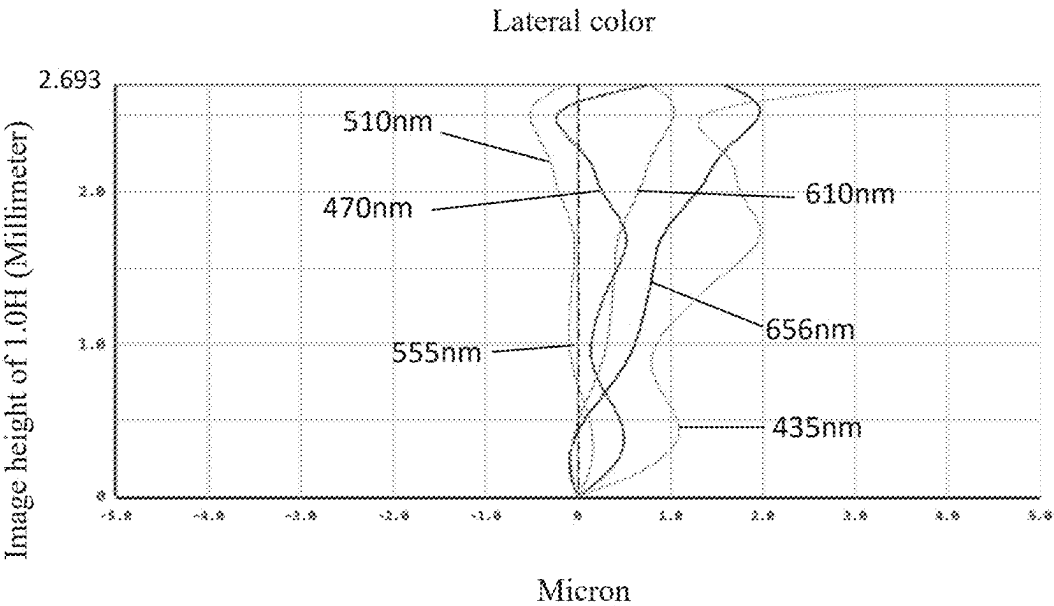


FIG. 7

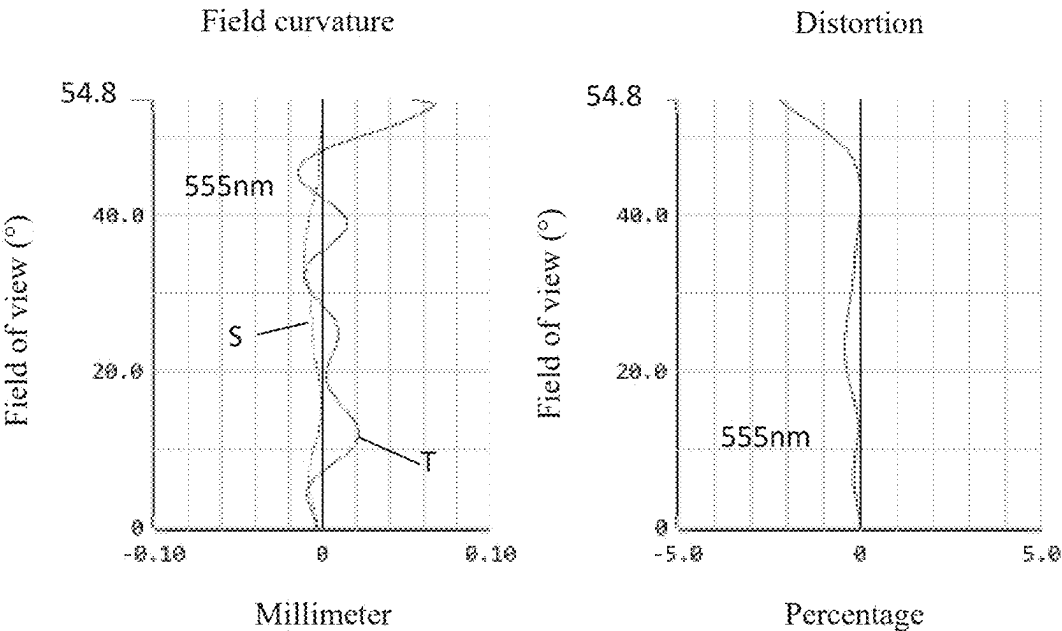


FIG. 8

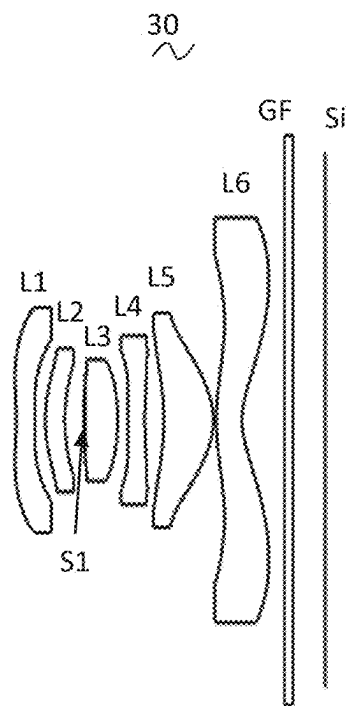


FIG. 9

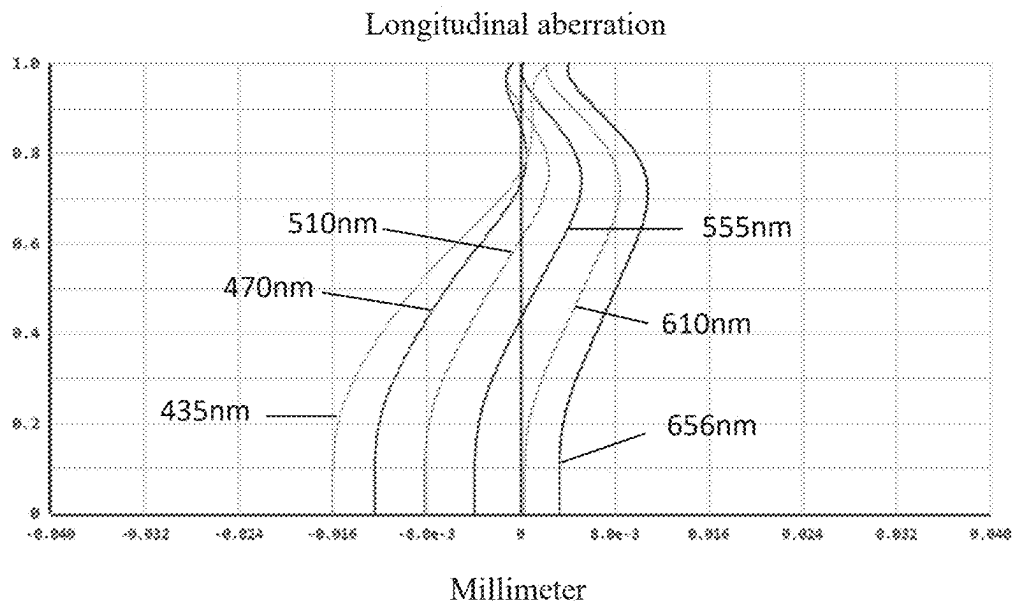


FIG. 10

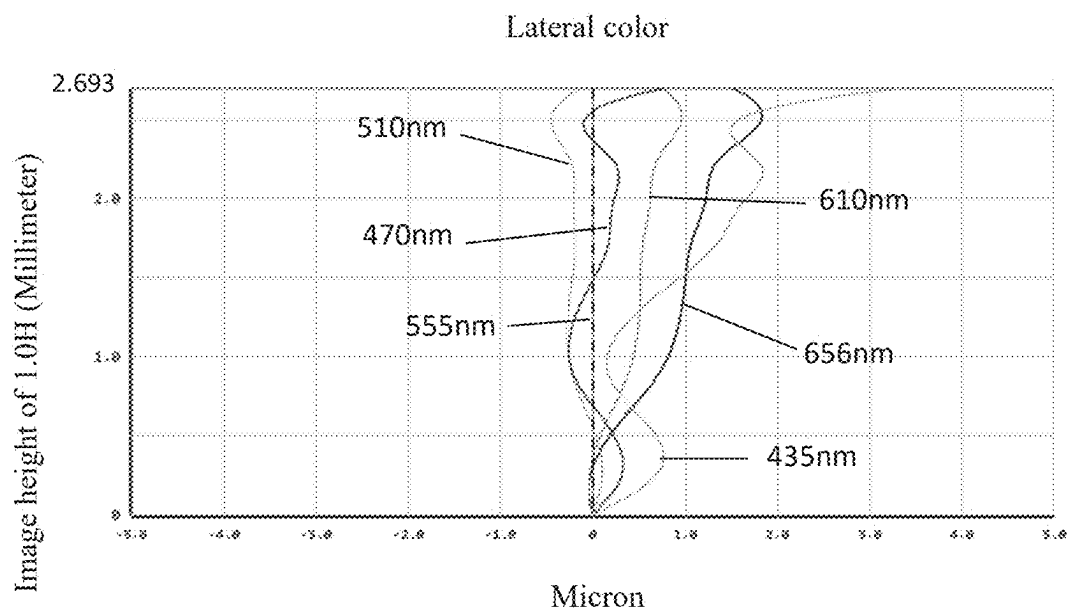


FIG. 11

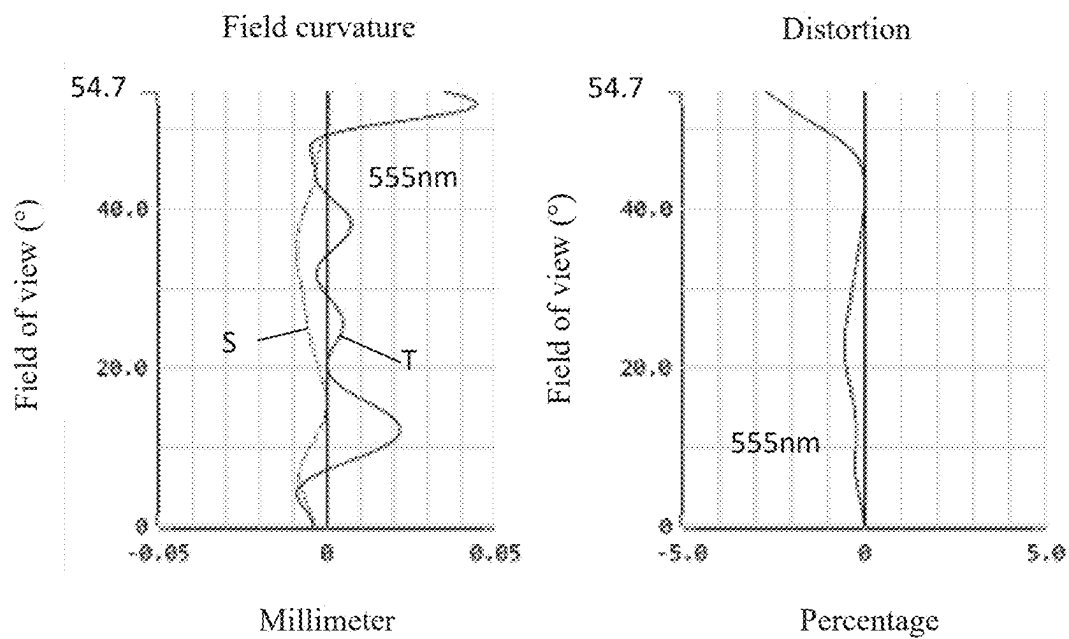


FIG. 12

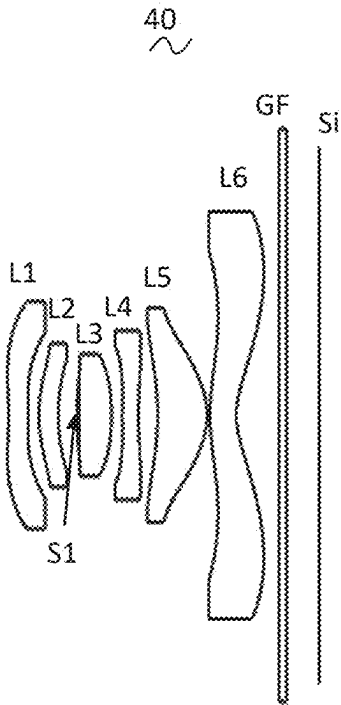


FIG. 13

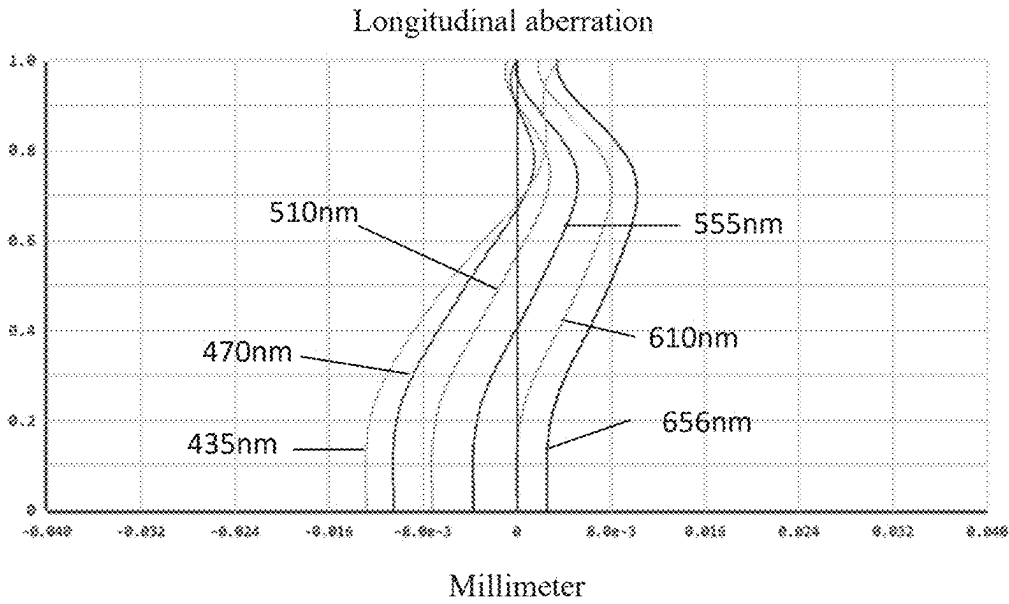


FIG. 14

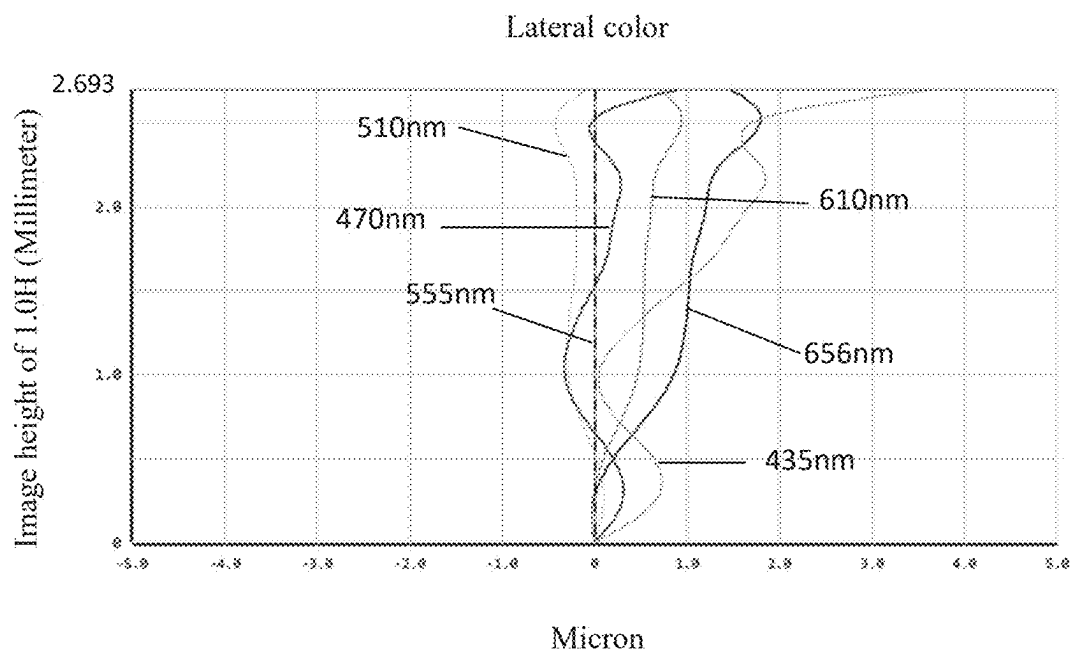


FIG. 15

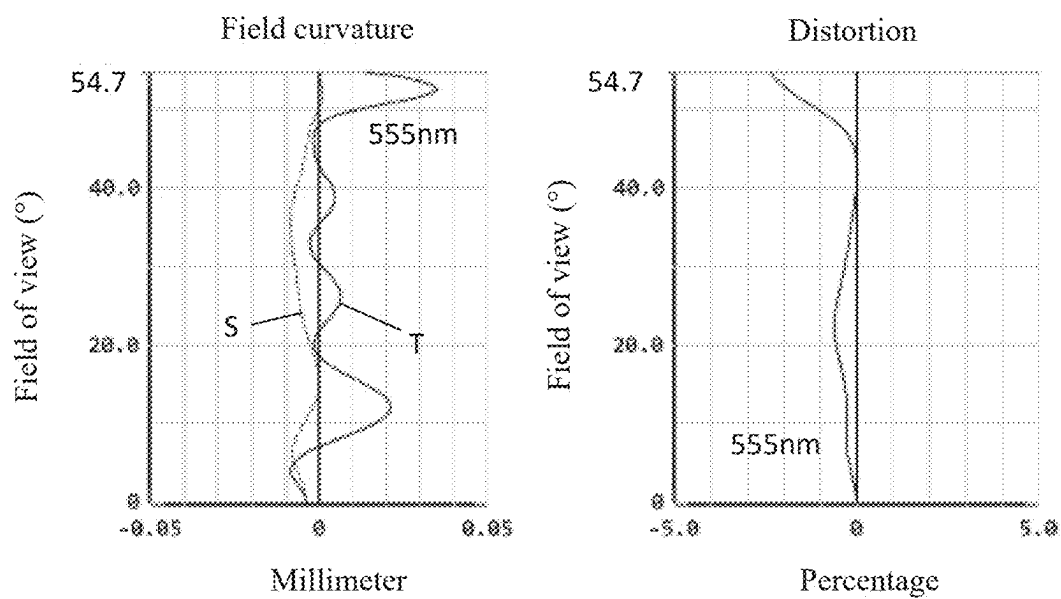


FIG. 16

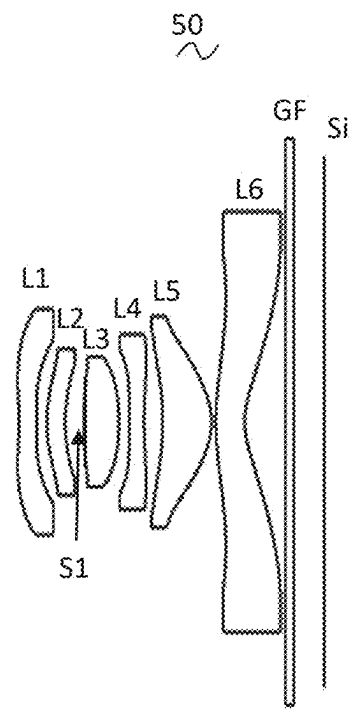


FIG. 17

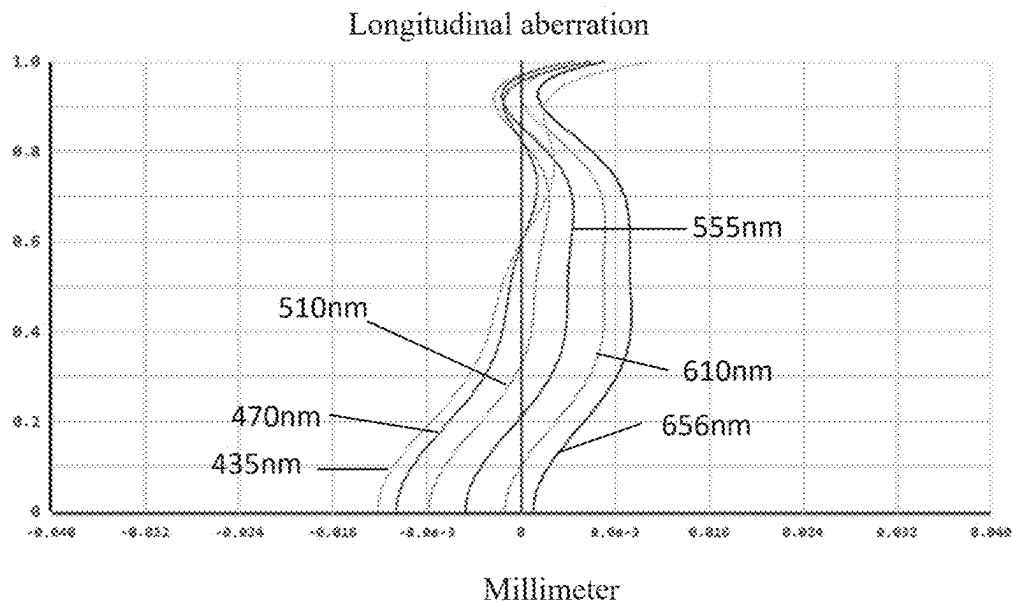


FIG. 18

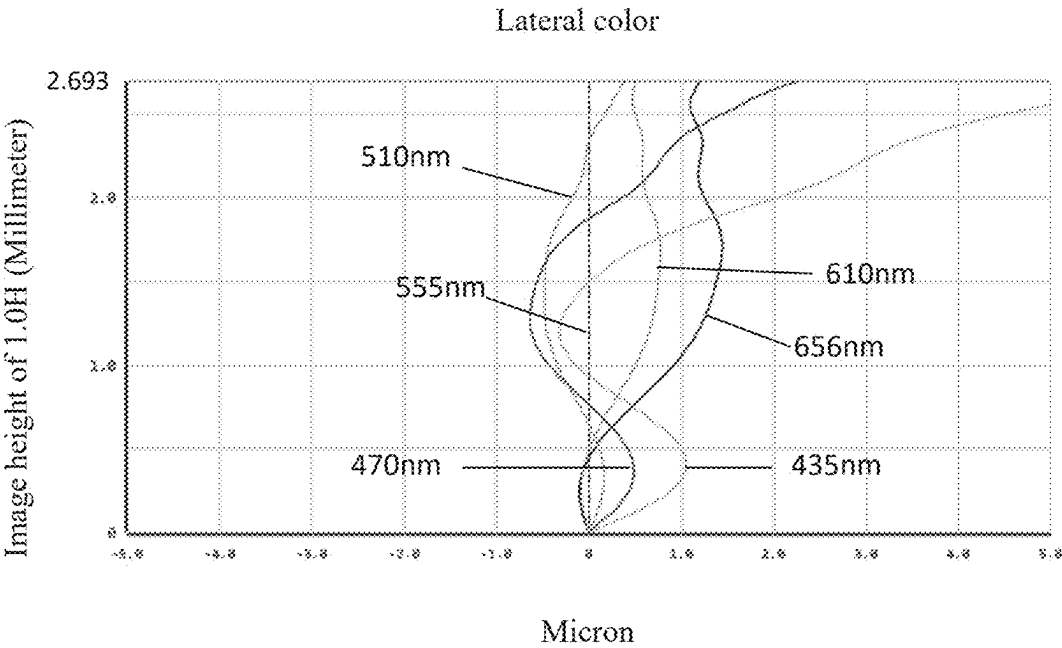


FIG. 19

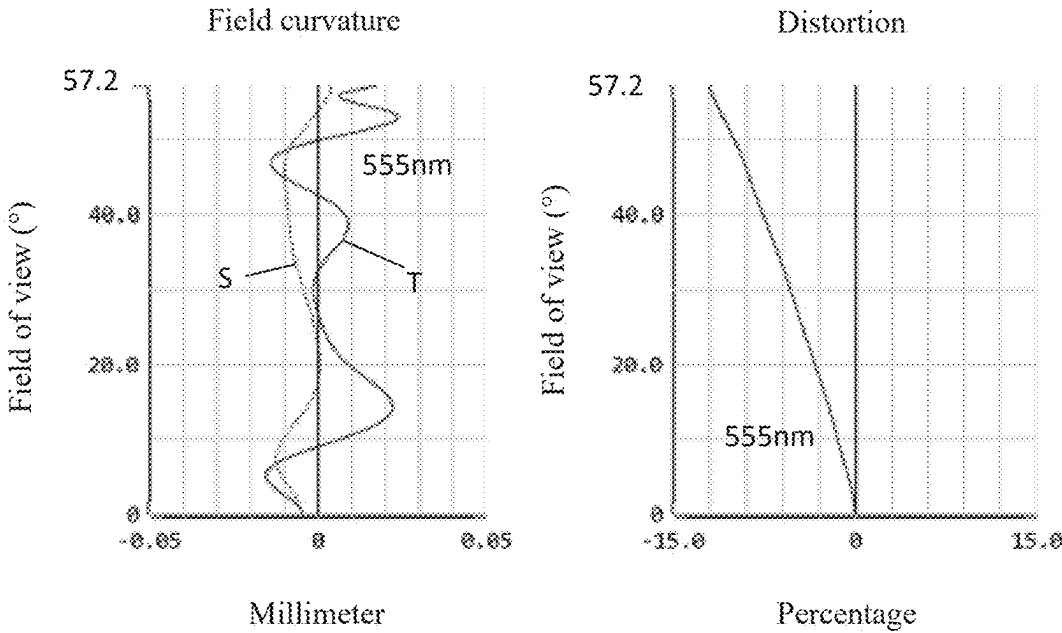


FIG. 20

CAMERA OPTICAL LENS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of priority under the Paris Convention to Chinese Patent Application No. 202410202516.8 filed on Feb. 23, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of optical lens, and in particular to a camera optical lens suitable for handheld devices such as smart phones and digital cameras and camera devices such as monitors and PC lenses.

BACKGROUND

[0003] With the emergence of smart devices in recent years, the demand for miniature camera optical lenses is increasing day by day, but the pixel size of the photosensitive devices is shrinking, coupled with the current development trend of electronic products being that their functions should be better and their shape should be thin and small, miniature camera optical lens with good imaging quality therefor has become a mainstream in the market. In order to obtain better imaging quality, the lenses traditionally adopt a multi-piece lens structure. And, with the development of technology and the increase of the diverse demands of users, and under this circumstances that the pixel area of photosensitive devices is shrinking steadily and the requirement of the system for the imaging quality is improving constantly, the six-piece lens structure gradually appear in lens design. There is an urgent need for wide-angle camera lenses which have good optical characteristics and small volume, and the chromatic aberration of which is fully corrected.

SUMMARY

[0004] Aiming at the above problem, the main purpose of the present disclosure is to provide a camera optical lens that has good optical performance and meets the design requirements of large aperture, ultra-thin, and wide-angle.

[0005] To this end, the technical solutions of the present disclosure provide a camera optical lens, including, from an object side to an image side in sequence, a first lens with a negative refractive power, a second lens with a positive refractive power, a third lens with a positive refractive power, a fourth lens with a negative refractive power, a fifth lens with a positive refractive power, and a sixth lens with a negative refractive power. The camera optical lens satisfies the following conditions: $6.00 \leq R2/R1 \leq 25.00$, $2.00 \leq (R7+R8)/(R7-R8) \leq 9.00$, and $3.00 \leq (d5+d9)/d7 \leq 9.00$, where R1 represents a curvature radius of an object side surface of the first lens, R2 represents a curvature radius of an image side surface of the first lens, R7 represents a curvature radius of an object side surface of the fourth lens, R8 represents a curvature radius of an image side surface of the fourth lens, d5 represents an on-axis thickness of the third lens, d7 represents an on-axis thickness of the fourth lens, and d9 represents an on-axis thickness of the fifth lens.

[0006] As an improvement, the camera optical lens further satisfies the following condition: $2.00 \leq TTL/f \leq 2.30$, where TTL represents a total track length of the camera optical lens, and f represents a focal length of the camera optical lens.

[0007] As an improvement, the camera optical lens further satisfies the following condition: $-6.00 \leq f6/d11 \leq -3.00$, where f6 represents a focal length of the sixth lens, and d11 represents an on-axis thickness of the sixth lens.

[0008] As an improvement, the camera optical lens further satisfies the following condition: $2.50 \leq f2/f3 \leq 7.00$, where f2 represents a focal length of the second lens, and f3 represents a focal length of the third lens.

[0009] As an improvement, the object side surface of the first lens is concave in a paraxial region, and the image side surface of the first lens is convex in the paraxial region. The camera optical lens further satisfies the following conditions: $-5.26 \leq f1/f \leq -1.23$, $-2.80 \leq (R1+R2)/(R1-R2) \leq -0.72$, and $0.02 \leq d1/TTL \leq 0.06$, where f1 represents a focal length of the first lens, f represents a focal length of the camera optical lens, d1 represents an on-axis thickness of the first lens, and TTL represents a total track length of the camera optical lens.

[0010] As an improvement, an object side surface of the second lens is convex in a paraxial region, and an image side surface of the second lens is concave in the paraxial region. The camera optical lens further satisfies the following conditions: $1.50 \leq f2/f \leq 11.95$, $-54.86 \leq (R3+R4)/(R3-R4) \leq -4.92$, and $0.03 \leq d3/TTL \leq 0.11$, where f2 represents a focal length of the second lens, f represents a focal length of the camera optical lens, R3 represents a curvature radius of the object side surface of the second lens, R4 represents a curvature radius of the image side surface of the second lens, d3 represents an on-axis thickness of the second lens, and TTL represents a total track length of the camera optical lens.

[0011] As an improvement, an object side surface of the third lens is convex in a paraxial region, and an image side surface of the third lens is convex in the paraxial region. The camera optical lens further satisfies the following conditions: $0.57 \leq f3/f \leq 1.79$, $0.11 \leq (R5+R6)/(R5-R6) \leq 0.37$, and $0.06 \leq d5/TTL \leq 0.18$, where f3 represents a focal length of the third lens, f represents a focal length of the camera optical lens, R5 represents a curvature radius of the object side surface of the third lens, R6 represents a curvature radius of an image side surface of the third lens, and TTL represents a total track length of the camera optical lens.

[0012] As an improvement, the object side surface of the fourth lens is convex in a paraxial region, and the image side surface of the fourth lens is concave in the paraxial region. The camera optical lens further satisfies the following conditions: $-9.47 \leq f4/f \leq -2.92$, and $0.03 \leq d7/TTL \leq 0.09$, where f4 represents a focal length of the fourth lens, f represents a focal length of the camera optical lens, and TTL represents a total track length of the camera optical lens.

[0013] As an improvement, an object side surface of the fifth lens is concave in a paraxial region, and an image side surface of the fifth lens is convex in the paraxial region. The camera optical lens further satisfies the following conditions: $0.29 \leq f5/f \leq 0.89$, $0.83 \leq (R9+R10)/(R9-R10) \leq 2.60$, and $0.09 \leq d9/TTL \leq 0.26$, where f5 represents a focal length of the fifth lens, f represents a focal length of the camera optical lens, R9 represents a curvature radius of the object side surface of the fifth lens, R10 represents a curvature radius of the image side surface of the fifth lens, and TTL represents a total track length of the camera optical lens.

[0014] As an improvement, an object side surface of the sixth lens is convex in a paraxial region, and an image side surface of the sixth lens is concave in the paraxial region. The camera optical lens further satisfies the following con-

ditions: $-1.44 \leq f_6/f \leq -0.48$, $0.89 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 2.74$, and $0.04 \leq d_{11}/TTL \leq 0.13$, where f_6 represents a focal length of the sixth lens, f represents a focal length of the camera optical lens, R_{11} represents a curvature radius of the object side surface of the sixth lens, R_{12} represents a curvature radius of the image side surface of the sixth lens, d_{11} represents an on-axis thickness of the sixth lens, and TTL represents a total track length of the camera optical lens.

[0015] The beneficial effects of the present disclosure are: the camera optical lens provided by the present disclosure has excellent optical performance and characteristics of large aperture, ultra-thin, and wide-angle, and is particularly suitable for mobile phone camera lens components and WEB camera lenses composed of camera elements such as charge coupled devices (CCD) or complementary metal-oxide semiconductor sensors (CMOS sensors) with high pixel.

BRIEF DESCRIPTION OF DRAWINGS

[0016] For clearer descriptions of the technical solutions in the embodiments of the present disclosure, drawings that are to be referred for description of the embodiments are briefly described hereinafter. Apparently, the drawings described hereinafter merely illustrate some embodiments of the present disclosure. Persons of ordinary skill in the art can derive other drawings based on the drawings described herein without any creative effort.

[0017] FIG. 1 is a schematic diagram of a structure of a camera optical lens according to a first embodiment of the present disclosure.

[0018] FIG. 2 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 1.

[0019] FIG. 3 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 1.

[0020] FIG. 4 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 1.

[0021] FIG. 5 is a schematic diagram of a structure of a camera optical lens according to a second embodiment of the present disclosure.

[0022] FIG. 6 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 5.

[0023] FIG. 7 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 5.

[0024] FIG. 8 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 5.

[0025] FIG. 9 is a schematic diagram of a structure of a camera optical lens according to a third embodiment of the present disclosure.

[0026] FIG. 10 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 9.

[0027] FIG. 11 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 9.

[0028] FIG. 12 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 9.

[0029] FIG. 13 is a schematic diagram of a structure of a camera optical lens according to a fourth embodiment of the present disclosure.

[0030] FIG. 14 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 13.

[0031] FIG. 15 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 13.

[0032] FIG. 16 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 13.

[0033] FIG. 17 is a schematic diagram of a structure of a camera optical lens according to a comparative example of the present disclosure.

[0034] FIG. 18 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 17.

[0035] FIG. 19 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 17.

[0036] FIG. 20 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 17.

DETAILED DESCRIPTION OF EMBODIMENTS

[0037] To make the objects, technical solutions, and advantages of the present disclosure clearer, embodiments of the present disclosure are described in detail with reference to accompanying drawings in the following. A person of ordinary skill in the art can understand that, in the embodiments of the present disclosure, many technical details are provided to make readers better understand the present disclosure. However, even without these technical details and any changes and modifications based on the following embodiments, technical solutions required to be protected by the present disclosure can be implemented.

[0038] Referring to FIGS. 1 to 16, the technical solutions of the present disclosure provide camera optical lenses 10, 20, 30, 40, and FIGS. 1, 5, 9 and 13 show the camera optical lenses 10, 20, 30, 40 according to the present disclosure, respectively. Each of the camera optical lenses 10, 20, 30, 40 includes six lenses. Specifically, the camera optical lens includes, from an object side to an image side: a first lens L1, a second lens L2, an aperture S1, a third lens L3, a fourth lens L4, a fifth lens L5, and a sixth lens L6. An optical element such as an optical filter (GF) may be arranged between the sixth lens L6 and an image surface Si.

[0039] The first lens L1, the second lens L2, the third lens L3, the fourth lens L4, the fifth lens L5 and the sixth lens L6 are all made of plastic material. In other embodiments, each lens may also be of another material.

[0040] A curvature radius of an object side surface of the first lens L1 is defined as R_1 , a curvature radius of an image side surface of the first lens L1 is defined as R_2 , and the camera optical lens satisfies a condition of $6.00 \leq R_2/R_1 \leq 25.00$, which stipulates a shape of the first lens L1. Within this range, astigmatism and distortion of the camera lens can be corrected, thereby reducing the possibility of dark corners.

[0041] A curvature radius of an object side surface of the fourth lens L4 is defined as R_7 , a curvature radius of an image side surface of the fourth lens L4 is defined as R_8 , and the camera optical lens satisfies a condition of $2.00 \leq (R_7+R_8)/(R_7-R_8) \leq 9.00$, which stipulates a shape of the fourth lens L4. Within this range, a degree of deflection of light passing through the lens can be alleviated, and aberrations can be reduced effectively.

[0042] An on-axis thickness of the third lens L3 is defined as d_5 , an on-axis thickness of the fourth lens L4 is defined as d_7 , an on-axis thickness of the fifth lens L5 is defined as d_9 , and the camera optical lens satisfies a condition of $3.00 \leq (d_5+d_9)/d_7 \leq 9.00$, which stipulates a ratio of a sum of the on-axis thickness of the third lens L3 and the on-axis thickness of the fifth lens L5 to the on-axis thickness of the fourth lens L4. Within this range, it is beneficial to reducing a total track length and thereby realizing an ultra-thin effect.

[0043] A total track length of the camera optical lens is defined as TTL , a focal length of the camera optical lens is defined as f , and the camera optical lens satisfies a condition

of $2.00 \leq \text{TTL}/f \leq 2.30$, which stipulates a ratio of the total track length to the focal length of the system. Within this range, it is beneficial for the system to allocate a respective reasonable thickness to each lens, thereby allocating refractive powers reasonably.

[0044] A focal length of the sixth lens L6 is defined as f_6 , an on-axis thickness of the sixth lens L6 is defined as d_{11} , and the camera optical lens satisfies a condition of $-6.00 \leq f_6/d_{11} \leq -3.00$, which stipulates a ratio of the focal length of the sixth lens L6 to the on-axis thickness of the sixth lens. Within this range, it is beneficial for balancing a thickness of the sixth lens, thereby allocating a refractive power of this lens reasonably.

[0045] A focal length of the second lens L2 is defined as f_2 , a focal length of the third lens L3 is defined as f_3 , and the camera optical lens satisfies a condition of $2.50 \leq f_2/f_3 \leq 7.00$, which stipulates a ratio of the focal length of the second lens L2 to the focal length of the third lens L3. Within this range, distribution of refractive powers can be balanced effectively.

[0046] When the above conditions are satisfied, the camera optical lenses 10, 20, 30, 40 can have excellent optical performance and meet design requirement of wide-angle. With the characteristics of the camera optical lenses 10, 20, 30, 40, they are particularly suitable for mobile phone camera lens components and WEB camera lenses composed of camera elements such as CCDs or CMOS sensors with high pixel.

[0047] Based on the above conditions and the functions that can be achieved, characteristics of each lens are illustrated in detail as follows.

[0048] The object side surface of the first lens L1 is concave in a paraxial region, the image side surface of the first lens L1 is convex in the paraxial region, and the first lens L1 has a negative refractive power. The object side surface and the image side surface of the first lens L1 may also be set to other concave or convex distribution situations.

[0049] A focal length of the camera optical lens is defined as f , a focal length of the first lens L1 is defined as f_1 , and the camera optical lens satisfies a condition of $-5.26 \leq f_1/f \leq -1.23$, which stipulates a ratio of the negative refractive power of the first lens L1 to the focal length of the camera optical lens. Within this range, the first lens L1 has a suitable negative refractive power, which is beneficial to reducing aberration of the system, and to development of the camera optical lens towards ultra-thin and wide-angle. As an improvement, the camera optical lens satisfies a condition of $-3.29 \leq f_1/f \leq -1.54$.

[0050] The curvature radius of the object side surface of the first lens L1 is defined as R_1 , the curvature radius of the image side surface of the first lens L1 is defined as R_2 , and the camera optical lens satisfies a condition of $-2.80 \leq (R_1 + R_2)/(R_1 - R_2) \leq -0.72$. In this way, the shape of the first lens L1 can be controlled reasonably, such that the first lens L1 can effectively correct a spherical aberration of the system. As an improvement, the camera optical lens satisfies a condition of $-1.75 \leq (R_1 + R_2)/(R_1 - R_2) \leq -0.90$.

[0051] An on-axis thickness of the first lens L1 is defined as d_1 , the total track length of the camera optical lens 10 is defined as TTL , and the camera optical lens satisfies a condition of $0.02 \leq d_1/\text{TTL} \leq 0.06$. Within this range, it is beneficial to implementing miniature camera optical lens. As an improvement, the camera optical lens satisfies a condition of $0.02 \leq d_1/\text{TTL} \leq 0.05$.

[0052] An object side surface of the second lens L2 is convex in a paraxial region, an image side surface of the second lens is concave in the paraxial region, and the second lens L2 has a positive refractive power. The object side surface and the image side surface of the second lens L2 may also be set to other concave or convex distribution situations.

[0053] The focal length of the camera optical lens is defined as f , the focal length of the second lens L2 is defined as f_2 , and the camera optical lens satisfies a condition of $1.50 \leq f_2/f \leq 11.95$. By controlling the positive refractive power of the second lens L2 within a reasonable range, it is beneficial to correcting an aberration of the optical system. As an improvement, the camera optical lens satisfies a condition of $2.40 \leq f_2/f \leq 9.56$.

[0054] A curvature radius of the object side surface of the second lens L2 is defined as R_3 , a curvature radius of the image side surface of the second lens L2 is defined as R_4 , and the camera optical lens satisfies a condition of $-54.86 \leq (R_3 + R_4)/(R_3 - R_4) \leq -4.92$, which stipulates a shape of the second lens L2. Within this range, a development towards ultra-thin and wide-angle lenses would facilitate correcting a problem of an on-axis aberration. As an improvement, the camera optical lens satisfies a condition of $-34.29 \leq (R_3 + R_4)/(R_3 - R_4) \leq -6.15$.

[0055] An on-axis thickness of the second lens L2 is defined as d_3 , the total track length of the camera optical lens 10 is defined as TTL , and the camera optical lens satisfies a condition of $0.03 \leq d_3/\text{TTL} \leq 0.11$. Within this range, it is beneficial to implementing miniature camera optical lens. As an improvement, the camera optical lens satisfies a condition of $0.05 \leq d_3/\text{TTL} \leq 0.09$.

[0056] An object side surface of the third lens L3 is convex in a paraxial region, an image side surface of the third lens is convex in the paraxial region, and the third lens L3 has a positive refractive power. The object side surface and the image side surface of the third lens L3 may also be set to other concave or convex distribution situations.

[0057] The focal length of the camera optical lens is defined as f , the focal length of the third lens L3 is defined as f_3 , and the camera optical lens satisfies a condition of $0.57 \leq f_3/f \leq 1.79$. By a reasonable allocation of the focal lengths, the system can have an excellent imaging quality and a lower sensitivity. As an improvement, the camera optical lens satisfies a condition of $0.92 \leq f_3/f \leq 1.44$.

[0058] A curvature radius of the object side surface of the third lens L3 is defined as R_5 , a curvature radius of an image side surface of the third lens L3 is defined as R_6 , and the camera optical lens satisfies a condition of $0.11 \leq (R_5 + R_6)/(R_5 - R_6) \leq 0.37$. Within this range, a shape of the third lens L3 can be effectively controlled, which is beneficial for the formation of the third lens L3 and to preventing poor formation and generation of stress due to excessive surface curvature of the third lens L3. As an improvement, the camera optical lens satisfies a condition of $0.18 \leq (R_5 + R_6)/(R_5 - R_6) \leq 0.29$.

[0059] The on-axis thickness of the third lens L3 is defined as d_5 , the total track length of the camera optical lens 10 is defined as TTL , and the camera optical lens satisfies a condition of $0.06 \leq d_5/\text{TTL} \leq 0.18$. Within this range, it is beneficial to implementing miniature camera optical lens. As an improvement, the camera optical lens satisfies a condition of $0.09 \leq d_5/\text{TTL} \leq 0.14$.

[0060] The object side surface of the fourth lens L4 is convex in a paraxial region, the image side surface of the

fourth lens L4 is concave in the paraxial region, and the fourth lens L4 has a negative refractive power. The object side surface and the image side surface of the fourth lens L4 may also be set to other concave or convex distribution situations.

[0061] The focal length of the camera optical lens is defined as f , a focal length of the fourth lens L4 is defined as f_4 , and the camera optical lens satisfies a condition of $-9.47 \leq f_4/f \leq -2.92$. By a reasonable allocation of the focal lengths, the system can have an excellent imaging quality and a lower sensitivity. As an improvement, the camera optical lens satisfies a condition of $-5.92 \leq f_4/f \leq -3.65$.

[0062] The on-axis thickness of the fourth lens L4 is defined as d_7 , the total track length of the camera optical lens 10 is defined as TTL, and the camera optical lens satisfies a condition of $0.03 \leq d_7/TTL \leq 0.09$. Within this range, it is beneficial to implementing miniature camera optical lens. As an improvement, the camera optical lens satisfies a condition of $0.04 \leq d_7/TTL \leq 0.07$.

[0063] An object side surface of the fifth lens L5 is concave in a paraxial region, an image side surface of the fifth lens is convex in the paraxial region, and the fifth lens L5 has a positive refractive power. The object side surface and the image side surface of the fifth lens L5 may also be set to other concave or convex distribution situations.

[0064] The focal length of the camera optical lens is defined as f , a focal length of the fifth lens L5 is defined as f_5 , and the camera optical lens satisfies a condition of $0.29 \leq f_5/f \leq 0.89$. By defining the fifth lens L5, a light angle for the camera optical lens 10 can be smoothed effectively and a tolerance sensitivity can be reduced. As an improvement, the camera optical lens satisfies a condition of $0.46 \leq f_5/f \leq 0.71$.

[0065] A curvature radius of the object side surface of the fifth lens L5 is defined as R_9 , a curvature radius of the image side surface of the fifth lens L5 is defined as R_{10} , and the camera optical lens satisfies a condition of $0.83 \leq (R_9 + R_{10})/(R_9 - R_{10}) \leq 2.60$, which stipulates a shape of the fifth lens L5. Within this range, a development towards ultra-thin and wide-angle lenses would facilitate correcting a problem of an off-axis aberration. As an improvement, the camera optical lens satisfies a condition of $1.33 \leq (R_9 + R_{10})/(R_9 - R_{10}) \leq 2.08$.

[0066] The on-axis thickness of the fifth lens L5 is defined as d_9 , the total track length of the camera optical lens 10 is defined as TTL, and the camera optical lens satisfies a condition of $0.09 \leq d_9/TTL \leq 0.26$. Within this range, it is beneficial to implementing miniature camera optical lens. As an improvement, the camera optical lens satisfies a condition of $0.14 \leq d_9/TTL \leq 0.21$.

[0067] An object side surface of the sixth lens L6 is convex in a paraxial region, an image side surface of the sixth lens is concave in the paraxial region, and the sixth lens L6 has a negative refractive power. The object side surface and the image side surface of the sixth lens L6 may also be set to other concave or convex distribution situations.

[0068] The focal length of the camera optical lens is defined as f , the focal length of the sixth lens L6 is defined as f_6 , and the camera optical lens satisfies a condition of $-1.44 \leq f_6/f \leq -0.48$. By a reasonable allocation of the focal lengths, the system can have an excellent imaging quality and a lower sensitivity. As an improvement, the camera optical lens satisfies a condition of $-0.90 \leq f_6/f \leq -0.59$.

[0069] A curvature radius of the object side surface of the sixth lens L6 is defined as R_{11} , a curvature radius of the image side surface of the sixth lens L6 is defined as R_{12} , and the camera optical lens satisfies a condition of $0.89 \leq (R_{11} + R_{12})/(R_{11} - R_{12}) \leq 2.74$, which stipulates a shape of the sixth lens L6. Within this range, a development towards ultra-thin and wide-angle lenses would facilitate correcting a problem of an off-axis aberration. As an improvement, the camera optical lens satisfies a condition of $1.42 \leq (R_{11} + R_{12})/(R_{11} - R_{12}) \leq 2.19$.

[0070] The on-axis thickness of the sixth lens L6 is defined as d_{11} , the total track length of the camera optical lens 10 is defined as TTL, and the camera optical lens satisfies a condition of $0.04 \leq d_{11}/TTL \leq 0.13$. Within this range, it is beneficial to implementing miniature camera optical lens. As an improvement, the camera optical lens satisfies a condition of $0.07 \leq d_{11}/TTL \leq 0.11$.

[0071] An image height of the camera optical lens is defined as IH, the total track length of the camera optical lens 10 is defined as TTL, and the camera optical lens satisfies a condition of $TTL/IH \leq 2.39$, which is beneficial to implementing ultra-thin. As an improvement, the camera optical lens satisfies a condition of $TTL/IH \leq 1.91$.

[0072] A field of view of the camera optical lens is defined as FOV, and the camera optical lens satisfies a condition of $FOV \geq 100.00^\circ$, it is facilitate to implementing wide-angle.

[0073] A F number of the camera optical lens is defined as FNO, and the camera optical lens satisfies a condition of $FNO \leq 2.26$. In this way, the camera optical lens can have a large aperture and a good imaging performance. As an improvement, the camera optical lens satisfies a condition of $FNO \leq 2.22$.

[0074] In the following, embodiments will be used to describe the camera optical lens of the present disclosure. The symbols recorded in each embodiment will be described as follows. The focal length, on-axis distance, curvature radius, on-axis thickness, inflexion point position, and arrest point position are all in units of mm.

[0075] TTL: total track length (the on-axis distance from the object side surface of the first lens L1 to the image surface Si of the camera optical lens along the optical axis) in mm.

[0076] The F number (FNO) refers to a ratio of an effective focal length of the camera optical lens to an entrance pupil diameter (ENPD).

[0077] Preferably, the object side surface and the image side surface of the lens may have inflexion points and/or arrest points, so as to satisfy the demand for high quality imaging.

[0078] In the following, the technical solutions of the present disclosure are illustrated in detail with reference to four embodiments, and at the same time, a comparative example is provided as a reference. The technical effect of the present disclosure cannot be achieved beyond the scope of the conditions as illustrated above.

First Embodiment

[0079] Table 1 and Table 2 show design data of the camera optical lens 10 according to the first embodiment of the present disclosure.

TABLE 1

	R	d	nd	vd
R1	-2.165	d1=	0.255 nd1	1.5444 v1
R2	-13.257	d2=	0.148	
R3	1.248	d3=	0.270 nd2	1.5876 v2
R4	1.400	d4=	0.283	
S1	∞	ds=	-0.024	
R5	3.116	d5=	0.471 nd3	1.5444 v3
R6	-1.959	d6=	0.144	
R7	6.254	d7=	0.233 nd4	1.6610 v4
R8	2.992	d8=	0.262	
R9	-2.156	d9=	0.688 nd5	1.5444 v5
R10	-0.545	d10=	0.040	
R11	1.891	d11=	0.354 nd6	1.6153 v6
R12	0.554	d12=	0.596	
R13	∞	d13=	0.110 ndg	1.5168 vg
R14	∞	d14=	0.448	

[0080] Herein, meanings of various symbols will be described as follows.

[0081] S1: aperture.

[0082] R: curvature radius of an optical surface, a central curvature radius for a lens.

[0083] R1: curvature radius of the object side surface of the first lens L1.

[0084] R2: curvature radius of the image side surface of the first lens L1.

[0085] R3: curvature radius of the object side surface of the second lens L2.

[0086] R4: curvature radius of the image side surface of the second lens L2.

[0087] R5: curvature radius of the object side surface of the third lens L3.

[0088] R6: curvature radius of the image side surface of the third lens L3.

[0089] R7: curvature radius of the object side surface of the fourth lens L4.

[0090] R8: curvature radius of the image side surface of the fourth lens L4.

[0091] R9: curvature radius of the object side surface of the fifth lens L5.

[0092] R10: curvature radius of the image side surface of the fifth lens L5.

[0093] R11: curvature radius of the object side surface of the sixth lens L6.

[0094] R12: curvature radius of the image side surface of the sixth lens L6.

[0095] R13: curvature radius of an object side surface of the optical filter (GF).

[0096] R14: curvature radius of an image side surface of the optical filter (GF).

[0097] d: on-axis thickness of a lens and an on-axis distance between lenses.

[0098] d1: on-axis thickness of the first lens L1.

[0099] d2: on-axis distance from the image side surface of the first lens L1 to the object side surface of the second lens L2.

[0100] d3: on-axis thickness of the second lens L2.

[0101] d4: on-axis distance from the image side surface of the second lens L2 to the aperture S1.

[0102] ds: on-axis distance from the aperture S1 to the object side surface of the third lens L3.

[0103] d5: on-axis thickness of the third lens L3.

[0104] d6: on-axis distance from the image side surface of the third lens L3 to the object side surface of the fourth lens L4.

[0105] d7: on-axis thickness of the fourth lens L4.

[0106] d8: on-axis distance from the image side surface of the fourth lens L4 to the object side surface of the fifth lens L5.

[0107] d9: on-axis thickness of the fifth lens L5.

[0108] d10: on-axis distance from the image side surface of the fifth lens L5 to the object side surface of the sixth lens L6.

[0109] d11: on-axis thickness of the sixth lens L6.

[0110] d12: on-axis distance from the image side surface of the sixth lens L6 to the object side surface of the optical filter (GF).

[0111] d13: on-axis thickness of the optical filter (GF).

[0112] d14: on-axis distance from the image side surface of the optical filter (GF) to the image surface Si.

[0113] nd: refractive index of a d line (the d line refers to green light with a wavelength of 550 nm).

[0114] nd1: refractive index of the d line of the first lens L1.

[0115] nd2: refractive index of the d line of the second lens L2.

[0116] nd3: refractive index of the d line of the third lens L3.

[0117] nd4: refractive index of the d line of the fourth lens L4.

[0118] nd5: refractive index of the d line of the fifth lens L5.

[0119] nd6: refractive index of the d line of the sixth lens L6.

[0120] ndg: refractive index of the d line of the optical filter (GF).

[0121] vd: abbe number.

[0122] v1: abbe number of the first lens L1.

[0123] v2: abbe number of the second lens L2.

[0124] v3: abbe number of the third lens L3.

[0125] v4: abbe number of the fourth lens L4.

[0126] v5: abbe number of the fifth lens L5.

[0127] V6: abbe number of the sixth lens L6.

[0128] vg: abbe number of the optical filter (GF).

[0129] Table 2 shows aspherical surface data of each lens of the camera optical lens 10 in the First Embodiment of the present disclosure.

TABLE 2

	Conic coefficient	Aspheric surface coefficients				
	k	A4	A6	A8	A10	A12
R1	-4.8154E+01	5.6403E-01	-8.7246E-01	1.0104E+00	-3.3790E-01	-1.0125E+00
R2	9.9000E+01	1.5489E+00	-5.4330E+00	2.0861E+01	-6.7077E+01	1.6223E+02

TABLE 2-continued

R3	-3.4155E+00	4.4573E-01	-3.2201E+00	1.4299E+01	-4.9880E+01	1.0997E+02
R4	3.6145E+00	-2.7705E-01	-3.6407E-01	-4.4214E+00	1.0130E+02	-1.0579E+03
R5	2.8836E+00	1.5098E-02	-9.1929E-01	1.9075E+01	-2.3092E+02	1.6794E+03
R6	1.2266E+00	-5.1375E-01	1.2845E+00	-4.3829E+00	4.5948E+00	5.9789E+01
R7	3.6232E+01	-8.5834E-01	2.2588E+00	-1.2601E+01	5.7063E+01	-1.5933E+02
R8	-1.3930E+00	-4.0698E-01	9.4368E-01	-3.0260E+00	9.5323E+00	-2.0727E+01
R9	-8.8345E-01	7.4188E-03	2.4574E-01	-6.2657E-01	1.4810E+00	2.2704E-01
R10	-1.1859E+00	1.4370E+00	-6.2369E+00	1.8818E+01	-3.8942E+01	5.3984E+01
R11	-3.7727E+00	-1.0219E-01	-2.4914E-01	6.3931E-01	-7.6787E-01	5.4699E-01
R12	-5.6596E+00	-7.5625E-02	-4.0208E-02	1.1131E-01	-1.0455E-01	5.5132E-02

	Conic coefficient	Aspheric surface coefficients			
	k	A14	A16	A18	A20
R1	-4.8154E+01	1.8855E+00	-1.5202E+00	6.1219E-01	-9.8521E-02
R2	9.9000E+01	-2.7619E+02	3.1315E+02	-2.1733E+02	7.1272E+01
R3	-3.4155E+00	-1.1459E+02	-4.7400E+01	2.3821E+02	-1.6440E+02
R4	3.6145E+00	6.1151E+03	-2.0237E+04	3.5867E+04	-2.6410E+04
R5	2.8836E+00	-7.4709E+03	1.9823E+04	-2.8724E+04	1.7396E+04
R6	1.2266E+00	-4.1769E+02	1.2621E+03	-1.9073E+03	1.1595E+03
R7	3.6232E+01	2.2336E+02	-4.3529E+01	-2.4931E+02	2.0503E+02
R8	-1.3930E+00	2.7261E+01	-1.9927E+01	6.7878E+00	-6.1089E-01
R9	-8.8345E-01	-6.4852E+00	1.0154E+01	-6.2871E+00	1.3716E+00
R10	-1.1859E+00	-4.8234E+01	2.6524E+01	-8.1686E+00	1.0795E+00
R11	-3.7727E+00	-2.4037E-01	6.3989E-02	-9.4742E-03	5.9937E-04
R12	-5.6596E+00	-1.7651E-02	3.4071E-03	-3.6589E-04	1.6840E-05

[0130] For convenience, an aspheric surface of each lens surface uses the aspheric surfaces as expressed in the following condition (1). However, the present disclosure is not limited to the aspherical polynomials form as expressed in the condition (1).

$$z = (cr^2)/\{1 + [1 - (k+1)(c^2r^2)]^{1/2}\} + A4r^4 + A6r^6 + A8r^8 + A10r^{10} + A12r^{12} + A14r^{14} + A16r^{16} + A18r^{18} + A20r^{20} \quad (1)$$

[0131] Herein, K is a conic coefficient, A4, A6, A8, A10, A12, A14, A16, A18, and A20 are aspheric surface coefficients, c is a curvature of an optical surface or a central curvature for a lens, r is a vertical distance between a point on an aspheric curve and the optical axis, and z is a depth of the aspheric surface (the vertical distance between a point on the aspheric surface from which a vertical distance to the optical axis is r and a tangent plane tangent to a vertex on the optical axis of the aspheric surface).

[0132] Table 3 and Table 4 show design data of inflexion points and arrest points of the lenses of the camera optical lens 10 according to the First Embodiment of the present disclosure. Herein P1R1 and P1R2 represent the object side surface and the image side surface of the first lens L1, P2R1 and P2R2 represent the object side surface and the image side surface of the second lens L2, P3R1 and P3R2 represent the object side surface and the image side surface of the third lens L3, P4R1 and P4R2 represent the object side surface and the image side surface of the fourth lens L4, P5R1 and P5R2 represent the object side surface and the image side surface of the fifth lens L5, P6R1 and P6R2 represent the object side surface and the image side surface of the sixth lens L6. The data in the column named “inflexion point position” refer to vertical distances from inflexion points arranged on surfaces of each lens to the optical axis of the camera optical lens 10. The data in the column named “arrest

point position” refer to vertical distances from arrest points arranged on surfaces of each lens to the optical axis of the camera optical lens 10.

TABLE 3

	Number of inflexion points	Inflexion point position 1	Inflexion point position 2	Inflexion point position 3
P1R1	1	0.215	/	/
P1R2	1	0.055	/	/
P2R1	3	0.515	0.715	0.725
P4R1	1	0.145	/	/
P4R2	1	0.345	/	/
P5R1	2	0.575	0.865	/
P5R2	1	0.765	/	/
P6R1	1	0.485	/	/
P6R2	1	0.475	/	/

TABLE 4

	Number of arrest points	Arrest point positions 1
P1R1	1	0.425
P1R2	1	0.085
P4R1	1	0.245
P4R2	1	0.755
P6R1	1	0.955
P6R2	1	1.355

[0133] FIG. 2 and FIG. 3 illustrate a longitudinal aberration and a lateral color of the camera optical lens 10 according to the First Embodiment after light with wavelengths of 656 nm, 610 nm, 555 nm, 510 nm, 470 nm and 435 nm passing through the camera optical lens 10, respectively. FIG. 4 illustrates a field curvature and a distortion of the camera optical lens 10 after light with a wavelength of

555 nm passing through the camera optical lens **10** according to the First Embodiment. A field curvature S in FIG. **4** is a field curvature in a sagittal direction, and T is a field curvature in a tangential direction.

[0134] In the embodiment, an entrance pupil diameter (ENPD) of the camera optical lens **10** is 0.897 mm, an image height IH of 1.0H is 2.693 mm, a field of view FOV in a diagonal direction is 111.80°. Thus, the camera optical lens can meet the design requirements of a large aperture, a wide angle and ultra-thin, and its on-axis and off-axis chromatic aberrations are fully corrected, thereby achieving excellent optical characteristics.

Second Embodiment

[0135] The symbols in the Second Embodiment have the same meanings as those in the First Embodiment.

TABLE 5-continued

	R	d	nd	vd
R7	6.837	d7=	0.237	nd4
R8	3.100	d8=	0.269	
R9	-2.192	d9=	0.696	nd5
R10	-0.543	d10=	0.040	
R11	1.888	d11=	0.347	nd6
R12	0.553	d12=	0.583	
R13	∞	d13=	0.110	ndg
R14	∞	d14=	0.434	

[0138] Table 6 shows aspherical surface data of each lens of the camera optical lens **20** in the Second Embodiment of the present disclosure.

TABLE 6

	Conic coefficient	Aspheric surface coefficients				
		k	A4	A6	A8	A10
R1	-3.8699E+01	5.7281E-01	-9.5623E-01	1.1697E+00	-4.4408E-01	-1.1286E+00
R2	9.9000E+01	1.4955E+00	-5.5489E+00	2.0609E+01	-6.0709E+01	1.2725E+02
R3	-3.7344E+00	3.7673E-01	-2.4688E+00	1.0184E+01	-3.9554E+01	1.1203E+02
R4	3.6053E+00	-2.9266E-01	4.9311E-01	-1.6355E+01	2.1080E+02	-1.7013E+03
R5	1.8398E+00	-1.3422E-02	3.5151E-01	-8.6970E+00	1.1835E+02	-1.0485E+03
R6	1.2207E+00	-4.4249E-01	8.8313E-01	-3.6727E+00	1.3721E+00	1.2724E+02
R7	3.8833E+01	-7.3300E-01	8.6403E-01	1.2975E-01	-2.8632E+01	2.2197E+02
R8	-2.3306E+00	-3.6759E-01	5.9764E-01	-1.3198E+00	2.5034E+00	6.3238E-01
R9	-2.4247E-01	1.1164E-03	3.3433E-01	-1.1659E+00	2.7016E+00	-1.8431E+00
R10	-1.2013E+00	1.4104E+00	-5.9620E+00	1.7421E+01	-3.4460E+01	4.5130E+01
R11	-3.5146E+00	-1.0636E-01	-1.6919E-01	4.7091E-01	-5.7291E-01	4.0906E-01
R12	-5.2189E+00	-6.0449E-02	-2.7468E-02	7.1627E-02	-6.4142E-02	3.2282E-02

	Conic coefficient	Aspheric surface coefficients			
		k	A14	A16	A18
R1	-3.8699E+01	2.1888E+00	-1.7825E+00	7.2140E-01	-1.1715E-01
R2	9.9000E+01	-1.7696E+02	1.5339E+02	-7.6806E+01	1.8321E+01
R3	-3.7344E+00	-2.0443E+02	2.0855E+02	-8.2495E+01	-8.2229E+00
R4	3.6053E+00	8.4059E+03	-2.4727E+04	3.9709E+04	-2.6790E+04
R5	1.8398E+00	5.9089E+03	-2.0364E+04	3.8914E+04	-3.1520E+04
R6	1.2207E+00	-9.2277E+02	3.0111E+03	-4.8436E+03	3.0991E+03
R7	3.8833E+01	-8.4940E+02	1.7868E+03	-1.9754E+03	8.9631E+02
R8	-2.3306E+00	-1.4940E+01	3.0525E+01	-2.6333E+01	8.5729E+00
R9	-2.4247E-01	-2.9791E+00	6.0484E+00	-3.8870E+00	8.6909E-01
R10	-1.2013E+00	-3.7671E+01	1.9131E+01	-5.3798E+00	6.4244E-01
R11	-3.5146E+00	-1.7899E-01	4.7179E-02	-6.8807E-03	4.2661E-04
R12	-5.2189E+00	-9.7692E-03	1.7573E-03	-1.7296E-04	7.1636E-06

[0136] FIG. **5** shows the camera optical lens **20** according to the Second Embodiment of the present disclosure.

[0137] Table 5 and Table 6 show design data of a camera optical lens **20** according to the Second Embodiment of the present disclosure.

TABLE 5

	R	d	nd	vd
R1	-1.879	d1=	0.229	nd1
R2	-46.974	d2=	0.121	
R3	1.078	d3=	0.281	nd2
R4	1.416	d4=	0.293	
S1	∞	ds=	-0.022	
R5	3.247	d5=	0.470	nd3
R6	-1.972	d6=	0.141	

TABLE 7

	Number of inflexion points	Inflexion point positions 1	Inflexion point positions 2
P1R1	1	0.225	/
P1R2	1	0.035	/
P2R1	2	0.505	0.715
P4R1	1	0.135	/
P4R2	1	0.325	/
P5R1	2	0.595	0.845
P5R2	2	0.765	1.135
P6R1	1	0.525	/
P6R2	1	0.505	/

TABLE 8

Number of arrest points		Arrest point positions 1
P1R1	1	0.455
P1R2	1	0.065
P4R1	1	0.235
P4R2	1	0.675
P6R1	1	1.085
P6R2	1	1.495

[0140] FIG. 6 and FIG. 7 illustrate a longitudinal aberration and a lateral color of the camera optical lens 20 according to the Second Embodiment after light with wavelengths of 656 nm, 610 nm, 555 nm, 510 nm, 470 nm and 435 nm passing through the camera optical lens 20, respectively. FIG. 8 illustrates a field curvature and a distortion of the camera optical lens 20 after light with a wavelength of 555 nm passing through the camera optical lens 20 according to the Second Embodiment. A field curvature S in FIG. 8 is a field curvature in a sagittal direction, and T is a field curvature in a tangential direction.

[0141] In the embodiment, an entrance pupil diameter (ENPD) of the camera optical lens 20 is 0.884 mm, an image height IH of 1.0H is 2.693 mm, a field of view FOV in a diagonal direction is 109.60°. Thus, the camera optical lens 20 can meet the design requirements of a large aperture, a wide angle and ultra-thin, and its on-axis and off-axis chromatic aberrations are fully corrected, thereby achieving excellent optical characteristics.

Third Embodiment

[0142] The symbols in the Third Embodiment have the same meanings as those in the First Embodiment.

[0143] FIG. 9 shows the camera optical lens 30 according to the Third Embodiment of the present disclosure.

[0144] Table 9 and Table 10 show design data of a camera optical lens 30 according to the Third Embodiment of the present disclosure.

TABLE 9

	R	d	nd	vd
R1	-2.165	d1=	0.255	nd1
R2	-13.257	d2=	0.148	1.5444 v1
R3	1.248	d3=	0.270	nd2
R4	1.400	d4=	0.283	1.5876 v2
S1	∞	ds=	-0.024	
R5	3.116	d5=	0.471	nd3
R6	-1.959	d6=	0.144	1.5444 v3
R7	6.254	d7=	0.233	nd4
R8	2.992	d8=	0.262	1.6610 v4
R9	-2.156	d9=	0.688	nd5
R10	-0.545	d10=	0.040	1.5444 v5
R11	1.891	d11=	0.354	nd6
R12	0.554	d12=	0.596	1.6153 v6
R13	∞	d13=	0.110	ndg
R14	∞	d14=	0.448	1.5168 vg

[0145] Table 10 shows aspherical surface data of each lens of the camera optical lens 30 in the Third Embodiment of the present disclosure.

TABLE 10

Conic coefficient		Aspheric surface coefficients				
	k	A4	A6	A8	A10	A12
R1	-4.0239E+01	5.6439E-01	-9.3316E-01	1.2703E+00	-1.0184E+00	1.1708E-01
R2	9.9000E+01	1.4550E+00	-4.7730E+00	1.7688E+01	-5.7108E+01	1.3824E+02
R3	-3.6099E+00	4.1849E-01	-2.7291E+00	1.1527E+01	-4.7927E+01	1.5281E+02
R4	3.5925E+00	-2.5157E-01	-2.0450E-01	-9.6116E+00	1.5884E+02	-1.4281E+03
R5	9.5113E-01	-7.1742E-03	-4.0194E-02	7.2471E-01	-2.3483E+01	2.4855E+02
R6	1.2115E+00	-4.4675E-01	1.0957E+00	-7.0937E+00	3.4398E+01	-8.4494E+01
R7	4.1136E+01	-7.5376E-01	1.3169E+00	-5.8100E+00	1.7299E+01	6.8679E-01
R8	-2.0892E+00	-3.7679E-01	7.0120E-01	-2.0202E+00	5.7024E+00	-8.9526E+00
R9	-4.3777E-01	1.5716E-02	2.7093E-01	-7.4895E-01	4.6357E-01	5.2311E+00
R10	-1.1849E+00	1.3774E+00	-5.8205E+00	1.7397E+01	-3.5730E+01	4.8954E+01
R11	-3.7456E+00	-1.2852E-01	-1.0935E-01	3.6722E-01	-4.8239E-01	3.7329E-01
R12	-5.2399E+00	-7.0655E-02	-1.4086E-02	5.9898E-02	-5.9662E-02	3.2949E-02

Conic coefficient		Aspheric surface coefficients			
	k	A14	A16	A18	A20
R1	-4.0239E+01	6.8322E-01	-7.2200E-01	3.1541E-01	-5.2028E-02
R2	9.9000E+01	-2.3029E+02	2.4641E+02	-1.5350E+02	4.3061E+01
R3	-3.6099E+00	-3.3719E+02	4.7185E+02	-3.6306E+02	1.1348E+02
R4	3.5925E+00	7.4919E+03	-2.2999E+04	3.8336E+04	-2.6838E+04
R5	9.5113E-01	-1.3817E+03	4.2318E+03	-6.7425E+03	4.3051E+03
R6	1.2115E+00	-5.2004E+01	8.2804E+02	-1.8210E+03	1.3348E+03
R7	4.1136E+01	-1.8645E+02	5.9080E+02	-7.8804E+02	3.9931E+02
R8	-2.0892E+00	3.3602E+00	9.3473E+00	-1.2872E+01	4.9712E+00
R9	-4.3777E-01	-1.6029E+01	1.9962E+01	-1.1807E+01	2.7249E+00
R10	-1.1849E+00	-4.3090E+01	2.3349E+01	-7.1151E+00	9.3748E-01
R11	-3.7456E+00	-1.7792E-01	5.1188E-02	-8.1462E-03	5.5023E-04
R12	-5.2399E+00	-1.0959E-02	2.1688E-03	-2.3480E-04	1.0695E-05

[0146] Table 11 and Table 12 show design data of inflexion points and arrest points of the lenses of the camera optical lens **30** according to the Third Embodiment of the present disclosure.

TABLE 11

	Number of inflexion points	Inflexion point positions 1	Inflexion point positions 2
P1R1	1	0.225	/
P1R2	1	0.075	/
P2R1	1	0.495	/
P4R1	1	0.145	/
P4R2	1	0.335	/
P5R1	2	0.595	0.855
P5R2	1	0.775	/
P6R1	1	0.515	/
P6R2	1	0.495	/

TABLE 12

	Number of arrest points	Arrest point positions 1
P1R1	1	0.435
P1R2	1	0.115
P4R1	1	0.245
P4R2	1	0.715
P6R1	1	1.025
P6R2	1	1.465

[0147] FIG. 10 and FIG. 11 illustrate a longitudinal aberration and a lateral color of the camera optical lens **30** according to the Third Embodiment after light with wavelengths of 656 nm, 610 nm, 555 nm, 510 nm, 470 nm and 435 nm passing through the camera optical lens **30**, respectively. FIG. 12 illustrates a field curvature and a distortion of the camera optical lens **30** after light with a wavelength of 555 nm passing through the camera optical lens **30** according to the Third Embodiment. A field curvature S in FIG. 12 is a field curvature in a sagittal direction, and T is a field curvature in a tangential direction.

[0148] In the embodiment, an entrance pupil diameter (ENPD) of the camera optical lens **30** is 0.890 mm, an image

height IH of 1.0H is 2.693 mm, a field of view FOV in a diagonal direction is 109.46°. Thus, the camera optical lens **30** can meet the design requirements of a large aperture, a wide angle and ultra-thin, and its on-axis and off-axis chromatic aberrations are fully corrected, thereby achieving excellent optical characteristics.

Fourth Embodiment

[0149] The symbols in the Fourth Embodiment have the same meanings as those in the First Embodiment.

[0150] FIG. 13 shows the camera optical lens **40** according to the Fourth Embodiment of the present disclosure.

[0151] Table 13 and Table 14 show design data of a camera optical lens **40** according to the Fourth Embodiment of the present disclosure.

TABLE 13

	R	d	nd	vd
R1	-2.312	d1=	0.265 nd1	1.5444 v1
R2	-13.875	d2=	0.155	
R3	1.295	d3=	0.265 nd2	1.5876 v2
R4	1.393	d4=	0.286	
S1	∞	ds=	-0.025	
R5	3.060	d5=	0.476 nd3	1.5444 v3
R6	-1.913	d6=	0.149	
R7	6.845	d7=	0.226 nd4	1.6610 v4
R8	3.064	d8=	0.265	
R9	-2.176	d9=	0.686 nd5	1.5444 v5
R10	-0.544	d10=	0.040	
R11	1.932	d11=	0.360 nd6	1.6153 v6
R12	0.553	d12=	0.592	
R13	∞	d13=	0.110 ndg	1.5168 vg
R14	∞	d14=	0.444	

[0152] Table 14 shows aspherical surface data of each lens of the camera optical lens **40** in the Fourth Embodiment of the present disclosure.

TABLE 14

	Conic coefficient	Aspheric surface coefficients				
	k	A4	A6	A8	A10	A12
R1	-4.0399E+01	5.6902E-01	-9.7449E-01	1.4548E+00	-1.5013E+00	9.0401E-01
R2	9.8984E+01	1.3992E+00	-4.2551E+00	1.4109E+01	-3.9833E+01	8.2819E+01
R3	-3.6672E+00	4.2617E-01	-2.7535E+00	1.1322E+01	-4.6461E+01	1.4744E+02
R4	3.5584E+00	-2.4014E-01	-1.7216E-01	-1.2878E+01	2.0438E+02	-1.7698E+03
R5	5.0334E-01	-7.4973E-03	1.3473E-02	-1.3802E+00	1.1067E+01	-8.8188E+01
R6	1.1748E+00	-4.3216E-01	1.1019E+00	-8.5763E+00	5.1299E+01	-1.9154E+02
R7	4.4318E+01	-7.4630E-01	1.4008E+00	-7.5968E+00	3.1433E+01	-6.4749E+01
R8	-2.2202E+00	-3.7914E-01	7.5313E-01	-2.6044E+00	8.8301E+00	-1.8739E+01
R9	-3.7971E-01	1.4964E-02	3.2042E-01	-1.3646E+00	3.9161E+00	-5.4760E+00
R10	-1.1832E+00	1.3663E+00	-5.7413E+00	1.7048E+01	-3.4735E+01	4.7147E+01
R11	-3.6955E+00	-1.2889E-01	-1.0207E-01	3.5348E-01	-4.6962E-01	3.6730E-01
R12	-5.2245E+00	-7.2200E-02	-2.8408E-03	4.1447E-02	-4.3219E-02	2.4196E-02

	Conic coefficient	Aspheric surface coefficients			
	k	A14	A16	A18	A20
R1	-4.0399E+01	-1.2745E-01	-2.0784E-01	1.3286E-01	-2.4412E-02
R2	9.8984E+01	-1.1652E+02	1.0323E+02	-5.2743E+01	1.2623E+01
R3	-3.6672E+00	-3.2782E+02	4.7132E+02	-3.8060E+02	1.2804E+02

TABLE 14-continued

R4	3.5584E+00	9.0087E+03	-2.6957E+04	4.3955E+04	-3.0193E+04
R5	5.0334E-01	6.0815E+02	-2.7728E+03	6.7355E+03	-6.5804E+03
R6	1.1748E+00	3.6065E+02	-1.2724E+02	-6.0535E+02	6.8369E+02
R7	4.4318E+01	1.6290E+00	2.6451E+02	-4.7715E+02	2.7556E+02
R8	-2.2202E+00	2.2104E+01	-1.2122E+01	5.5756E-01	1.4592E+00
R9	-3.7971E-01	3.2522E+00	-9.5578E-02	-6.1462E-01	1.2830E-01
R10	-1.1832E+00	-4.1057E+01	2.1982E+01	-6.6126E+00	8.5958E-01
R11	-3.6955E+00	-1.7680E-01	5.1306E-02	-8.2256E-03	5.5917E-04
R12	-5.2245E+00	-8.0976E-03	1.6058E-03	-1.7362E-04	7.8762E-06

[0153] Table 15 and Table 16 show design data of inflexion points and arrest points of the lenses of the camera optical lens **40** according to the Fourth Embodiment of the present disclosure.

TABLE 15

	Number of inflexion points	Inflexion point positions 1	Inflexion point positions 2	Inflexion point positions 3
P1R1	1	0.225	/	/
P1R2	1	0.075	/	/
P2R1	3	0.485	0.725	0.745
P4R1	1	0.135	/	/
P4R2	1	0.325	/	/
P5R1	2	0.595	0.855	/
P5R2	1	0.775	/	/
P6R1	1	0.515	/	/
P6R2	1	0.505	/	/

TABLE 16

	Number of arrest points	Arrest point positions 1
P1R1	1	0.435
P1R2	1	0.115
P4R1	1	0.235
P4R2	1	0.685
P6R1	1	1.025
P6R2	1	1.495

[0154] FIG. 14 and FIG. 15 illustrate a longitudinal aberration and a lateral color of the camera optical lens **40** according to the Fourth Embodiment after light with wavelengths of 656 nm, 610 nm, 555 nm, 510 nm, 470 nm and 435 nm passing through the camera optical lens **40**, respectively. FIG. 16 illustrates a field curvature and a distortion of the camera optical lens **40** after light with a wavelength of 555 nm passing through the camera optical lens **40** according to the Fourth Embodiment. A field curvature S in FIG. 16 is a field curvature in a sagittal direction, and T is a field curvature in a tangential direction.

[0155] In the embodiment, an entrance pupil diameter (ENPD) of the camera optical lens **40** is 0.887 mm, an image height IH of 1.0H is 2.693 mm, a field of view FOV in a diagonal direction is 109.47°. Thus, the camera optical lens **40** can meet the design requirements of a large aperture, a wide angle and ultra-thin, and its on-axis and off-axis chromatic aberrations are fully corrected, thereby achieving excellent optical characteristics.

[0156] Table 21 in the following shows various values in the First Embodiment, Second Embodiment, Third Embodiment and Fourth Embodiment, and also values corresponding to parameters which are specified in the above conditions.

Comparative Example

[0157] The symbols in the Comparative Example have the same meanings as those in the First Embodiment.

[0158] FIG. 17 shows the camera optical lens **50** according to the Comparative Example of the present disclosure.

[0159] Table 17 and Table 18 show design data of a camera optical lens **50** according to the Comparative Example of the present disclosure.

TABLE 17

	R	d	nd	vd
R1	-2.318	d1=	0.244	nd1
R2	-13.911	d2=	0.142	
R3	1.300	d3=	0.255	nd2
R4	1.394	d4=	0.277	
S1	∞	ds=	-0.022	
R5	3.070	d5=	0.487	nd3
R6	-1.875	d6=	0.140	
R7	7.532	d7=	0.209	nd4
R8	3.223	d8=	0.258	
R9	-2.185	d9=	0.675	nd5
R10	-0.535	d10=	0.061	
R11	2.352	d11=	0.395	nd6
R12	0.544	d12=	0.569	
R13	∞	d13=	0.110	ndg
R14	∞	d14=	0.421	

[0160] Table 18 shows aspherical surface data of each lens of the camera optical lens **50** in the Comparative Example of the present disclosure.

TABLE 18

	Conic coefficient	Aspheric surface coefficients				
		k	A4	A6	A8	A10
R1	-3.7062E+01	5.2264E-01	-8.0838E-01	9.4080E-01	-4.0079E-01	-6.6408E-01
R2	9.9000E+01	1.3718E+00	-4.8341E+00	2.0017E+01	-6.7893E+01	1.6284E+02
R3	-3.8512E+00	4.2002E-01	-2.8459E+00	1.3006E+01	-5.4625E+01	1.5839E+02
R4	3.5272E+00	-2.2376E-01	-3.3990E-01	-9.1214E+00	1.5304E+02	-1.3636E+03
R5	-5.7336E-01	-9.8893E-03	2.1883E-01	-5.6612E+00	4.7322E+01	-2.3617E+02
R6	1.3606E+00	-4.0508E-01	3.8307E-01	1.4696E+00	-3.4073E+01	2.4385E+02

TABLE 18-continued

R7	4.6865E+01	-7.5707E-01	1.6409E+00	-1.0573E+01	4.9847E+01	-1.4093E+02
R8	-2.6313E+00	-3.8099E-01	8.8245E-01	-3.5279E+00	1.2261E+01	-2.7002E+01
R9	-2.3847E-01	5.1325E-02	-2.7528E-01	2.3865E+00	-9.1334E+00	2.2238E+01
R10	-1.1780E+00	1.2708E+00	-5.0149E+00	1.4417E+01	-2.9086E+01	3.9647E+01
R11	-1.9253E+00	-1.4676E-01	1.4898E-01	-1.9924E-01	1.9838E-01	-1.2810E-01
R12	-5.3278E+00	-7.1994E-02	6.3693E-02	-5.8529E-02	3.8971E-02	-1.7027E-02
Conic coefficient		Aspheric surface coefficients				
	k	A14	A16	A18	A20	
R1	-3.7062E+01	1.3122E+00	-1.0232E+00	3.9290E-01	-6.0525E-02	
R2	9.9000E+01	-2.6124E+02	2.6618E+02	-1.5680E+02	4.1277E+01	
R3	-3.8512E+00	-3.0382E+02	3.7651E+02	-2.6955E+02	8.2954E+01	
R4	3.5272E+00	6.9240E+03	-2.0245E+04	3.1882E+04	-2.1024E+04	
R5	-5.7336E-01	6.5610E+02	-7.7713E+02	-3.9535E+02	1.3270E+03	
R6	1.3606E+00	-9.8019E+02	2.2738E+03	-2.8316E+03	1.4525E+03	
R7	4.6865E+01	2.1510E+02	-1.2045E+02	-7.1721E+01	8.7535E+01	
R8	-2.6313E+00	3.5397E+01	-2.5864E+01	8.8322E+00	-7.5124E-01	
R9	-2.3847E-01	-3.3818E+01	3.0525E+01	-1.4892E+01	2.9952E+00	
R10	-1.1780E+00	-3.5187E+01	1.9592E+01	-6.2785E+00	8.9036E-01	
R11	-1.9253E+00	5.2232E-02	-1.2978E-02	1.7945E-03	-1.0589E-04	
R12	-5.3278E+00	4.7205E-03	-7.9775E-04	7.4888E-05	-2.9911E-06	

[0161] Table 19 and Table 20 show design data of inflexion points and arrest points of the lenses of the camera optical lens **50** according to the Comparative Example of the present disclosure.

TABLE 19

	Number of inflexion points	Inflexion point positions 1	Inflexion point positions 2	Inflexion point positions 3
P1R1	1	0.235	/	/
P1R2	1	0.075	/	/
P2R1	2	0.485	0.715	/
P3R1	1	0.475	/	/
P4R1	2	0.135	0.755	/
P4R2	1	0.315	/	/
P5R1	2	0.605	0.855	/
P5R2	1	0.795	/	/
P6R1	1	0.635	/	/
P6R2	3	0.545	2.145	2.215

TABLE 20

	Number of arrest points	Arrest point positions 1
P1R1	1	0.455
P1R2	1	0.115
P4R1	1	0.225
P4R2	1	0.645

TABLE 20-continued

	Number of arrest points	Arrest point positions 1
P6R1	1	1.315
P6R2	1	1.975

[0162] FIG. 18 and FIG. 19 illustrate a longitudinal aberration and a lateral color of the camera optical lens **50** according to the Comparative Example after light with wavelengths of 656 nm, 610 nm, 555 nm, 510 nm, 470 nm and 435 nm passing through the camera optical lens **50**, respectively. FIG. 20 illustrates a field curvature and a distortion of the camera optical lens **50** after light with a wavelength of 555 nm passing through the camera optical lens **50** according to the Comparative Example. A field curvature S in FIG. 20 is a field curvature in a sagittal direction, and T is a field curvature in a tangential direction. [0163] Table 21 in the following also shows various values in the Comparative Example corresponding to parameters in the above conditions. Obviously, the camera optical lens **50** according to the Comparative Example does not satisfy the above condition $2.50 \leq f2/f3 \leq 7.00$.

[0164] In the Comparative Example, an entrance pupil diameter (ENPD) of the camera optical lens **50** is 0.896 mm, an image height IH of 1.0H is 2.693 mm, a field of view FOV in a diagonal direction is 114.39°. Thus, the camera optical lens **50** cannot meet the design requirements of a wide angle.

TABLE 21

Parameters and Conditions	First Embodiment	Second Embodiment	Third Embodiment	Fourth Embodiment	Comparative Example
(d5 + d9)/d7	4.74	4.92	4.97	5.14	5.55
(R7 + R8)/(R7 - R8)	3.08	2.66	2.83	2.62	2.50
R2/R1	10.52	25.00	6.12	6.00	6.00
f	1.972	1.940	1.954	1.948	1.969
f1	-4.385	-3.590	-4.777	-5.122	-5.132
f2	9.195	5.830	11.723	15.515	16.294
f3	2.321	2.320	2.277	2.230	2.208
f4	-9.338	-8.727	-8.856	-8.521	-8.616
f5	1.139	1.150	1.159	1.155	1.134

TABLE 21-continued

Parameters and Conditions	First Embodimen	Second Embodiment	Third Embodimen	Fourth Embodimen	Comparative Example
f6	-1.409	-1.401	-1.407	-1.390	-1.248
FNO	2.20	2.20	2.20	2.20	2.20
TTL	4.017	3.999	4.025	4.029	4.220
IH	2.693	2.693	2.693	2.693	2.693
FOV	111.80	109.60	109.46	109.47	114.39

[0165] Those skilled in the art shall understand that the embodiments described above are specific embodiments for implementing the present disclosure. In practice, various changes may be made to these embodiments in form and in detail without departing from the spirit and scope of the disclosure.

What is claimed is:

1. A camera optical lens, comprising, from an object side to an image side in sequence, a first lens with a negative refractive power, a second lens with a positive refractive power, a third lens with a positive refractive power, a fourth lens with a negative refractive power, a fifth lens with a positive refractive power, and a sixth lens with a negative refractive power;

wherein the camera optical lens satisfies the following conditions:

$$6.00 \leq R2/R1 \leq 25.00;$$

$$2.00 \leq (R7 + R8)/(R7 - R8) \leq 9.00; \text{ and}$$

$$3.00 \leq (d5 + d9)/d7 \leq 9.00;$$

wherein

R1 represents a curvature radius of an object side surface of the first lens;

R2 represents a curvature radius of an image side surface of the first lens;

R7 represents a curvature radius of an object side surface of the fourth lens;

R8 represents a curvature radius of an image side surface of the fourth lens;

d5 represents an on-axis thickness of the third lens;

d7 represents an on-axis thickness of the fourth lens; and

d9 represents an on-axis thickness of the fifth lens.

2. The camera optical lens according to claim 1, further satisfies the following condition:

$$2.00 \leq TTL/f \leq 2.30;$$

wherein

TTL represents a total track length of the camera optical lens; and

f represents a focal length of the camera optical lens.

3. The camera optical lens according to claim 1, further satisfies the following condition:

$$-6.00 \leq f6/d11 \leq -3.00;$$

wherein

f6 represents a focal length of the sixth lens; and

d11 represents an on-axis thickness of the sixth lens.

4. The camera optical lens according to claim 1, further satisfies the following condition:

$$2.50 \leq f2/f3 \leq 7.00;$$

wherein

f2 represents a focal length of the second lens; and

f3 represents a focal length of the third lens.

5. The camera optical lens according to claim 1, wherein the object side surface of the first lens is concave in a paraxial region, and the image side surface of the first lens is convex in the paraxial region;

wherein the camera optical lens further satisfies the following conditions:

$$-5.26 \leq f1/f \leq -1.23;$$

$$-2.80 \leq (R1 + R2)/(R1 - R2) \leq -0.72; \text{ and}$$

$$0.02 \leq d1/TTL \leq 0.06;$$

wherein

f1 represents a focal length of the first lens;

f represents a focal length of the camera optical lens;

d1 represents an on-axis thickness of the first lens; and

TTL represents a total track length of the camera optical lens.

6. The camera optical lens according to claim 1, wherein an object side surface of the second lens is convex in a paraxial region, and an image side surface of the second lens is concave in the paraxial region;

wherein the camera optical lens further satisfies the following conditions:

$$1.50 \leq f2/f \leq 11.95;$$

$$-54.86 \leq (R3 + R4)/(R3 - R4) \leq -4.92; \text{ and}$$

$$0.03 \leq d3/TTL \leq 0.11;$$

wherein

f2 represents a focal length of the second lens;

f represents a focal length of the camera optical lens;

R3 represents a curvature radius of the object side surface of the second lens;

R4 represents a curvature radius of the image side surface of the second lens;

d3 represents an on-axis thickness of the second lens; and TTL represents a total track length of the camera optical lens.

7. The camera optical lens according to claim 1, wherein an object side surface of the third lens is convex in a paraxial region, and an image side surface of the third lens is convex in the paraxial region;

wherein the camera optical lens further satisfies the following conditions:

$$\begin{aligned} 0.57 &\leq f3/f \leq 1.79; \\ 0.11 &\leq (R5 + R6)/(R5 - R6) \leq 0.37; \text{ and} \\ 0.06 &\leq d5/TTL \leq 0.18; \end{aligned}$$

wherein

f3 represents a focal length of the third lens;

f represents a focal length of the camera optical lens;

R5 represents a curvature radius of the object side surface of the third lens;

R6 represents a curvature radius of an image side surface of the third lens; and

TTL represents a total track length of the camera optical lens.

8. The camera optical lens according to claim 1, wherein the object side surface of the fourth lens is convex in a paraxial region, and the image side surface of the fourth lens is concave in the paraxial region;

wherein the camera optical lens further satisfies the following conditions:

$$\begin{aligned} -9.47 &\leq f4/f \leq -2.92; \text{ and} \\ 0.03 &\leq d7/TTL \leq 0.09; \end{aligned}$$

wherein

f4 represents a focal length of the fourth lens;

f represents a focal length of the camera optical lens; and

TTL represents a total track length of the camera optical lens.

9. The camera optical lens according to claim 1, wherein an object side surface of the fifth lens is concave in a paraxial region, and an image side surface of the fifth lens is convex in the paraxial region;

wherein the camera optical lens further satisfies the following conditions:

$$\begin{aligned} 0.29 &\leq f5/f \leq 0.89; \\ 0.83 &\leq (R9 + R10)/(R9 - R10) \leq 2.60; \text{ and} \\ 0.09 &\leq d9/TTL \leq 0.26; \end{aligned}$$

wherein

f5 represents a focal length of the fifth lens;

f represents a focal length of the camera optical lens;

R9 represents a curvature radius of the object side surface of the fifth lens;

R10 represents a curvature radius of the image side surface of the fifth lens; and

TTL represents a total track length of the camera optical lens.

10. The camera optical lens according to claim 1, wherein an object side surface of the sixth lens is convex in a paraxial region, and an image side surface of the sixth lens is concave in the paraxial region;

wherein the camera optical lens further satisfies the following conditions:

$$\begin{aligned} -1.44 &\leq f6/f \leq -0.48; \\ 0.89 &\leq (R11 + R12)/(R11 - R12) \leq 2.74; \text{ and} \\ 0.04 &\leq d11/TTL \leq 0.13; \end{aligned}$$

wherein

f6 represents a focal length of the sixth lens;

f represents a focal length of the camera optical lens;

R11 represents a curvature radius of the object side surface of the sixth lens;

R12 represents a curvature radius of the image side surface of the sixth lens;

d11 represents an on-axis thickness of the sixth lens; and

TTL represents a total track length of the camera optical lens.

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